

Mathematics

Basic 9

Complete BECE Preparation
Examination Papers & Marking Schemes

Beacon Educational Consult

2026 Edition

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Published by Beacon Educational Consult
Accra, Ghana

First Edition: 2026

ISBN: [To be assigned]

Printed in Ghana

This book has been prepared in accordance with the
Ghana Education Service Standards-Based Curriculum (2019)
and the National Council for Curriculum and Assessment (NaCCA) guidelines.

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Preface

This comprehensive Mathematics textbook for Basic 9 has been carefully prepared to support Junior High School students in their preparation for the Basic Education Certificate Examination (BECE).

The book contains:

- Multiple choice questions with complete answer keys
- Theory questions with detailed marking schemes
- Curriculum-aligned content following the 2019 Standards-Based Curriculum
- Practice papers from BECE Mock examinations

Each question has been verified for accuracy and aligned with the appropriate curriculum objectives. The marking schemes provide detailed explanations to help students understand the reasoning behind each answer.

We hope this book will be a valuable resource in your academic journey.

— Beacon Educational Consult

Accra, Ghana

2026

BEACON EDUCATIONAL SERIES

Mathematics for Basic 9

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 9 (JHS 3) Students and Teachers

Beacon Educational Consult
2026 Edition

Mathematics for Basic 9
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

Published by: Beacon Educational Consult

ISBN: [To be assigned]

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PREFACE

This book has been developed to support Junior High School Basic 9 (JHS 3) students and teachers in their study of Mathematics, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY

Chapter 1: Ch10

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Solve $x + 3 > 5$.
A. $x < 2$
B. $x > 2$
C. $x > 8$
D. $x > -2$
2. Solve $2x \leq 8$.
A. $x \leq 16$
B. $x < 4$
C. $x \leq 4$
D. $x \geq 4$
3. Solve $x - 4 < 1$.
A. $x < 5$
B. $x < 4$
C. $x > 5$
D. $x < -3$
4. Solve $3x > 12$.
A. $x < 4$
B. $x > 9$
C. $x > 36$
D. $x > 4$
5. Solve $-x < 3$.
A. $x > -3$
B. $x > 3$
C. $x < 3$
D. $x < -3$
6. Solve $-2x \geq 6$.
A. $x \leq 3$
B. $x \leq -3$
C. $x \geq -3$
D. $x \geq 3$
7. Solve $x + \frac{1}{2} \geq -\frac{3}{2}$.
A. $x \leq -2$
B. $x \geq -1$
C. $x \geq -2$
D. $x \geq 2$
8. Solve $3x - 2 \geq 7$.
A. $x \geq 3$
B. $x \geq 5$
C. $x \geq 2$
D. $x \leq 3$
9. Solve $5 - x > 2$.
A. $x > -3$
B. $x < 3$
C. $x > 3$
D. $x < -3$
10. Solve $x/3 < 2$.
A. $x < 6$
B. $x > 6$
C. $x < 2/3$
D. $x < 3/2$
11. Solve $2x/3 \geq 4$.
A. $x \geq 2$
B. $x < 6$
C. $x \geq 6$
D. $x \leq 6$
12. Solve $4 - 3x \leq 10$.
A. $x \leq 2$
B. $x \leq -2$
C. $x \geq 2$
D. $x \geq -2$
13. Solve $5x + 2 < 3x + 10$.
A. $x > 4$
B. $x < 4$
C. $x < 6$
D. $x > 8$
14. Solve $3(x - 1) \geq 2x + 1$.
A. $x \geq 4$
B. $x \geq 2$
C. $x \geq -4$
D. $x \leq 4$

15. Solve $-(2/3)x + 3 \geq 3$.

- A. $x \geq 3$
- B. $x \geq 0$
- C. $x \leq 0$
- D. $x \leq 3$

17. The inequality shown on the number line is

- A. $x < 3$
- B. $x \leq 3$
- C. $x \geq 3$
- D. $x > 3$

19. An open (unshaded) circle on a number-line graph shows that

- A. the end value is included in the solution
- B. the solution is zero
- C. the arrow points left
- D. the end value is NOT included in the solution

21. The inequality shown on the number line is

- A. $x > 4$
- B. $x < 4$
- C. $x \geq 4$
- D. $x \leq 4$

23. The solution of $-3x - 8 > -26$ is (curriculum exemplar)

- A. $x < -6$
- B. $x > -6$
- C. $x < 6$
- D. $x > 6$

25. The solution of $2x \geq 8$ is (curriculum exemplar)

- A. $x \leq 4$
- B. $x > 4$
- C. $x \geq 16$
- D. $x \geq 4$

27. The solution of $7x/3 < 7$ is (curriculum exemplar)

- A. $x < 21$
- B. $x < 3$
- C. $x > 3$
- D. $x < 1$

16. Solve $\frac{1}{2}(5x - 5) < x + 5$.

- A. $x < 5$
- B. $x > 5$
- C. $x < 15$
- D. $x < 10$

18. The inequality shown on the number line is

- A. $x \leq 2$
- B. $x < -2$
- C. $x \leq -2$
- D. $x \geq -2$

20. A closed (shaded) circle on a number-line graph shows that

- A. there is no solution
- B. all numbers solve it
- C. the end value is included in the solution
- D. the end value is not included

22. Which graph represents $x < 5$?

- A. open circle at 5 with arrow to the right
- B. open circle at 5 with arrow to the left
- C. closed circle at 5 with arrow to the left
- D. closed circle at 5 with arrow to the right

24. The solution of $2x - 3 \leq 19$ is (curriculum exemplar)

- A. $x \leq 11$
- B. $x < 11$
- C. $x \geq 11$
- D. $x \leq 22$

26. The solution of $5y + 3 < 6 + 2y$ is (curriculum exemplar)

- A. $y < 1$
- B. $y < 3$
- C. $y > 3$
- D. $y > 1$

28. The solution set $x < 6$ is illustrated on the number line with

- A. an open circle at 6 and an arrow to the left
- B. an open circle at 6 and an arrow to the right
- C. a closed circle at 6 and an arrow to the left
- D. a closed circle at -6 and an

arrow to the left

29. The statement 'x is at least 7' is written as

- A. $x \leq 7$
- B. $x < 7$

- C. $x > 7$
D. $x \geq 7$
- 31.** Ama has GH¢50. She buys books at GH¢5 each and wants to keep at least GH¢20. The number n of books she can buy satisfies
A. $n \geq 6$
B. $n \leq 10$
C. $n < 6$
D. $n \leq 6$
- 33.** Kofi's age a is at least 12 years. This is written as
A. $a < 12$
B. $a > 12$
C. $a \geq 12$
D. $a \leq 12$
- 35.** To pass, a student needs at least 45 marks. She has 30 marks so far. The mark t she must score in the final test satisfies
A. $t > 75$
B. $t \geq 45$
C. $t \leq 15$
D. $t \geq 15$
- 37.** The sum of three consecutive integers is less than 30. The largest possible value of the smallest integer is
A. 9
B. 10
C. 8
D. 27
- 39.** The values of x that satisfy both $x > 2$ and $x < 5$ are
A. $x > 2$
B. $x < 2$
C. $2 < x < 5$
D. $x > 5$
- 30.** The statement ' x is at most 7' is written as
A. $x > 7$
B. $x \leq 7$
C. $x < 7$
D. $x \geq 7$
- 32.** Twice a number, increased by 5, is greater than 17. The number satisfies
A. $x > 12$
B. $x > 6$
C. $x > 11$
D. $x < 6$
- 34.** A taxi charges GH¢3 per trip plus GH¢2 per kilometre. With at most GH¢23, the number of kilometres k satisfies
A. $k \leq 10$
B. $k \geq 10$
C. $k \leq 13$
D. $k \leq 20$
- 36.** A rectangle has width 4 cm and a perimeter of at most 24 cm. Its length l satisfies
A. $l \leq 20$
B. $l \leq 8$
C. $l \geq 8$
D. $l \leq 10$
- 38.** The solution of $\frac{1}{2}(2x + 3) \leq x + 2$ is
A. no solution
B. $x \geq 2$
C. $x \leq 2$
D. all real numbers (always true)
- 40.** The smallest integer that satisfies $x \geq -1$ is
A. 0
B. -1
C. 1
D. -2

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Solving simple inequalities

Question 2

- (a) Inequalities with fractions and brackets

Question 3

- (a) Illustrating solutions on the number line

Question 4

- (a) Reading and explaining graphs

Question 5

- (a) Real-life problems

Question 6

- (a) Create and solve

Chapter 2: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Angles at a point add up to
 - A. 90°
 - B. 180°
 - C. 360°
 - D. 270°
2. Angles on a straight line add up to
 - A. 180°
 - B. 360°
 - C. 270°
 - D. 90°
3. Vertically opposite angles are
 - A. right angles
 - B. supplementary
 - C. equal
 - D. complementary
4. Find the value of x.
 - A. 70°
 - B. 110°
 - C. 20°
 - D. 290°
5. Find the value of x.
 - A. 65°
 - B. 25°
 - C. 130°
 - D. 115°
6. Find the value of x.
 - A. 95°
 - B. 215°
 - C. 120°
 - D. 145°
7. The 'F-shape' angles formed by a transversal cutting parallel lines are called
 - A. alternate angles
 - B. co-interior angles
 - C. corresponding angles
 - D. vertically opposite angles
8. The 'Z-shape' angles formed by a transversal cutting parallel lines are called
 - A. co-interior angles
 - B. vertically opposite angles
 - C. corresponding angles
 - D. alternate angles
9. Co-interior ('C-shape') angles between parallel lines are
 - A. reflex
 - B. supplementary
 - C. equal
 - D. complementary
10. Lines a and b are parallel. Find x.
 - A. 70°
 - B. 100°
 - C. 110°
 - D. 120°
11. Lines a and b are parallel. Find y.
 - A. 70°
 - B. 20°
 - C. 55°
 - D. 110°
12. The complement of 35° is
 - A. 35°
 - B. 145°
 - C. 55°
 - D. 65°
13. A polygon with five sides is called a
 - A. hexagon
 - B. quadrilateral
 - C. heptagon

D. pentagon

15. The sum of the interior angles of a triangle is

- A. 360°
- B. 90°
- C. 270°
- D. 180°

14. A polygon with six sides is called a

- A. hexagon
- B. octagon
- C. nonagon
- D. pentagon

- 17.** The sum of interior angles of a pentagon is
A. 720°
B. 900°
C. 360°
D. 540°
- 19.** The sum of interior angles of an octagon is
A. 720°
B. $1,080^\circ$
C. $1,260^\circ$
D. $1,440^\circ$
- 21.** Each interior angle of a regular pentagon is
A. 144°
B. 72°
C. 120°
D. 108°
- 23.** Each exterior angle of a regular octagon is
A. 30°
B. 45°
C. 90°
D. 40°
- 25.** The exterior angle is 120° . Find x .
A. 60°
B. 75°
C. 120°
D. 45°
- 27.** Congruent triangles have corresponding sides that are
A. equal
B. perpendicular
C. proportional
D. parallel
- 29.** The triangles are similar. Find x .
A. 12
B. 8
C. 4.5
D. 6
- 16.** Using $(n - 2) \times 180^\circ$, the sum of interior angles of a quadrilateral is
A. 180°
B. 360°
C. 540°
D. 720°
- 18.** The sum of interior angles of a hexagon is
A. 720°
B. $1,080^\circ$
C. 540°
D. 900°
- 20.** A polygon whose interior angles sum to 900° has
A. 7 sides
B. 8 sides
C. 9 sides
D. 6 sides
- 22.** Each interior angle of a regular hexagon is
A. 120°
B. 108°
C. 135°
D. 60°
- 24.** Find x in the quadrilateral.
A. 100°
B. 80°
C. 95°
D. 90°
- 26.** Find x in the hexagon.
A. 120°
B. 130°
C. 110°
D. 100°
- 28.** Similar triangles have corresponding angles that are
A. proportional
B. supplementary
C. equal
D. complementary
- 30.** Two similar triangles have corresponding sides 5 cm and 15 cm. A side of 7 cm in the smaller triangle corresponds to
A. 21 cm
B. 3 cm

- C. 7 cm
- D. 14 cm

32. The RHS congruence condition applies to

- A. any triangles
- B. obtuse triangles only
- C. equilateral triangles only
- D. right-angled triangles

31. Which condition does NOT guarantee congruent triangles?

- A. SSS
- B. AAA
- C. SAS
- D. ASA

- 33.** Two similar triangles have sides in the ratio 2 : 3. If the smaller triangle's perimeter is 20 cm, the larger's is
- A. 45 cm
 - B. 13.3 cm
 - C. 30 cm
 - D. 40 cm
- 35.** The angle between the hands of a clock at 3 o'clock is
- A. 60°
 - B. 90°
 - C. 180°
 - D. 30°
- 37.** Four equal angles at a point each measure
- A. 45°
 - B. 90°
 - C. 60°
 - D. 120°
- 39.** Each exterior angle of a regular polygon is 30° . The polygon has
- A. 30 sides
 - B. 10 sides
 - C. 6 sides
 - D. 12 sides
- 34.** Two similar triangles have sides in the ratio 2 : 3. The ratio of their areas is
- A. 2 : 3
 - B. 8 : 27
 - C. 6 : 9
 - D. 4 : 9
- 36.** Two angles on a straight line are $2x$ and $3x$. The value of x is
- A. 60
 - B. 30
 - C. 36
 - D. 45
- 38.** An isosceles triangle has a vertex angle of 40° . Each base angle is
- A. 70°
 - B. 40°
 - C. 55°
 - D. 140°
- 40.** The interior angles of a polygon sum to $1,440^\circ$. The polygon is a
- A. octagon
 - B. nonagon
 - C. decagon
 - D. dodecagon

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Angle properties at a point, on a line and opposite angles

Question 2

(a) Parallel lines and a transversal

Question 3

(a) The polygon angle-sum formula

Question 4

(a) Missing angles in figures

Question 5

(a) Similar and congruent triangles

Question 6

(a) Create and solve

Chapter 3: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The point where the three angle bisectors of a triangle meet is called the
A. circumcentre
B. centroid
C. orthocentre
D. incentre
2. The incentre of a triangle is equidistant from
A. the midpoints of the three sides
B. the two longest sides
C. the three sides
D. the three vertices
3. To locate the incentre of a triangle by construction, the lines drawn are
A. the bisectors of any two angles
B. the three altitudes
C. the perpendicular bisectors of any two sides
D. the three medians
4. The radius of the inscribed circle of a triangle equals
A. the distance from the incentre to any vertex
B. half of the longest side
C. the perpendicular distance from the incentre to any side
D. a third of the perimeter
5. The point where the perpendicular bisectors of the sides of a triangle meet is called the
A. centroid
B. circumcentre
C. midpoint
D. incentre
6. The circumcentre of a triangle is equidistant from
A. the incentre
B. the three sides
C. the three vertices
D. the midpoints of the sides
7. The circle that passes through all three vertices of a triangle is its
A. inscribed circle
B. circumscribed circle
C. semicircle
D. unit circle
8. The perpendicular bisector of segment AB is the locus of points
A. equidistant from A and B
B. on the segment AB
C. equidistant from two parallel lines
D. 2 cm from A
9. The bisector of $\angle ABC$ is the locus of points
A. midway between A and C
B. equidistant from B and C
C. equidistant from the arms BA and BC
D. a fixed distance from A
10. The minimum number of angle bisectors needed to locate the incentre of a triangle is
A. 0
B. 1
C. 3
D. 2

- 11.** The minimum number of perpendicular bisectors needed to locate the circumcentre of a triangle is
- 3
 - 0
 - 2
 - 1
- 13.** In triangle ABC, I is the incentre and the perpendicular distances from I to each side are 3 cm. The radius of the inscribed circle is
- 1.5 cm
 - 3 cm
 - 6 cm
 - 9 cm
- 15.** The circumcentre of a right-angled triangle lies at
- a point outside the triangle
 - the midpoint of the hypotenuse
 - the right-angle vertex
 - the incentre
- 17.** The circumcentre lies OUTSIDE the triangle when the triangle is
- right-angled
 - obtuse-angled
 - equilateral
 - acute-angled
- 19.** A square ABCD is constructed with $AB = 6.5$ cm. Measuring the diagonal AC gives about
- 4.6 cm
 - 9.2 cm
 - 13.0 cm
 - 6.5 cm
- 21.** Each side of the square in question 20 is approximately
- 16 cm
 - 11.3 cm
 - 5.7 cm
 - 4.0 cm
- 23.** A rectangle has length 6 cm and diagonal 10 cm. Its width is
- 12 cm
 - 16 cm
 - 4 cm
 - 8 cm
- 12.** The instruments used to construct the perpendicular bisector of a line segment are
- a set square and a protractor
 - a protractor and a ruler
 - a ruler only
 - a pair of compasses and a ruler
- 14.** S is the centre of the circle passing through P, Q and R. SP is equal to
- half of PQ
 - the radius of the inscribed circle
 - SQ
 - the perpendicular distance from S to PQ
- 16.** The circumradius of a right-angled triangle whose hypotenuse is 10 cm is
- 5 cm
 - 20 cm
 - 10 cm
 - 2.5 cm
- 18.** In an equilateral triangle, the incentre and the circumcentre
- are the same point
 - both lie outside the triangle
 - lie on different sides
 - are the ends of one side
- 20.** To construct a square from a given diagonal of 8 cm, one should first
- bisect the 8 cm line into four equal parts
 - construct a 60° angle at one end
 - draw a side of 8 cm
 - draw the 8 cm diagonal and construct its perpendicular bisector
- 22.** The minimum data needed to construct a rectangle are
- one side only
 - the lengths of two adjacent sides
 - one diagonal only
 - one side and one angle
- 24.** The diagonal of a rectangle with sides 8 cm and 15 cm is
- 17 cm
 - 15 cm
 - 23 cm
 - 12 cm

- 25.** The property used when constructing a rhombus with a given side is that a rhombus has
- A. equal diagonals
 - B. all angles equal
 - C. only one pair of parallel sides
 - D. all four sides equal
- 27.** A parallelogram can be constructed uniquely when given
- A. one angle only
 - B. two adjacent sides and the included angle
 - C. one side only
 - D. one diagonal only
- 29.** Which of the following sets of lengths CANNOT form a triangle?
- A. 5 cm, 12 cm, 13 cm
 - B. 3 cm, 4 cm, 8 cm
 - C. 6 cm, 8 cm, 10 cm
 - D. 3 cm, 4 cm, 5 cm
- 31.** Constructing triangle ABC with $AB = 5$ cm, $BC = 6$ cm and $\angle ABC = 60^\circ$ uses the
- A. SSS condition
 - B. SAS condition
 - C. ASA condition
 - D. RHS condition
- 33.** When constructing a regular hexagon inside a circle of radius r , each side of the hexagon equals
- A. πr
 - B. $2r$
 - C. $r/2$
 - D. r
- 35.** A square of side 4 cm has an equilateral triangle attached on top. The perimeter of the compound shape is
- A. 24 cm
 - B. 28 cm
 - C. 20 cm
 - D. 16 cm
- 37.** The locus of a point equidistant from two fixed points P and Q is
- A. a circle centred at P
 - B. the line PQ
 - C. the bisector of angle POQ
- 26.** A rhombus has diagonals 6 cm and 8 cm. Each side of the rhombus is
- A. 5 cm
 - B. 10 cm
 - C. 14 cm
 - D. 7 cm
- 28.** Using the SSS condition with sides 5 cm, 6 cm and 7 cm, the number of different (non-congruent) triangles that can be constructed is
- A. 2
 - B. 3
 - C. infinitely many
 - D. 1
- 30.** A triangle has sides 5 cm and 8 cm, with the third side a whole number x cm. The least possible value of x is
- A. 13
 - B. 2
 - C. 4
 - D. 3
- 32.** Constructing a triangle with angles 60° and 60° and the included side 4 cm uses the
- A. ASA condition
 - B. SAS condition
 - C. RHS condition
 - D. SSS condition
- 34.** A regular hexagon is constructed inside a circle of radius 4 cm. The perimeter of the hexagon is
- A. 24 cm
 - B. 8 cm
 - C. 16 cm
 - D. 32 cm
- 36.** The locus of a point at a fixed distance of 4 cm from a fixed point O is
- A. a circle of radius 4 cm about O
 - B. the perpendicular bisector of two points
 - C. a straight line
 - D. a pair of parallel lines

D. the perpendicular bisector of PQ

- 39.** Arcs of equal radius from A and B meet at X and Y, and XY bisects AB at right angles. For the construction to work, the radius must be
- A. exactly half of AB
 - B. less than half of AB
 - C. equal to AB
 - D. greater than half of AB
- 38.** In constructing the circumscribed circle of a triangle, the line drawn to bisect each side is its
- A. median
 - B. altitude
 - C. perpendicular bisector
 - D. angle bisector
- 40.** A circle drawn inside a square of side 6 cm touches all four sides. The diameter of the circle is
- A. 6 cm
 - B. 3 cm
 - C. 12 cm
 - D. 9 cm

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Constructing a triangle and its inscribed circle

Question 2

- (a) Constructing a triangle and its circumscribed circle

Question 3

- (a) Constructing a square

Question 4

- (a) Constructing a rectangle from a side and a diagonal

Question 5

- (a) Parallelogram and rhombus

Question 6

- (a) Equilateral triangle: inscribed and circumscribed circles

Chapter 4: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The cross-section of a prism is
 - A. the same shape and size along its length
 - B. a circle that widens along the length
 - C. a shape that changes along the length
 - D. a single point at each end
2. The number of faces of a cuboid is
 - A. 8
 - B. 6
 - C. 12
 - D. 4
3. The number of vertices of a cuboid is
 - A. 8
 - B. 12
 - C. 6
 - D. 10
4. The number of edges of solid P is
 - A. 6
 - B. 8
 - C. 12
 - D. 9
5. Solid Q is a
 - A. pyramid
 - B. cone
 - C. cuboid
 - D. triangular prism
6. The net of a cuboid is made up of
 - A. 4 rectangles and 2 squares
 - B. 6 squares
 - C. 6 rectangles
 - D. 4 rectangles and 2 triangles
7. The net of a triangular prism is made up of
 - A. 3 triangles and 2 rectangles
 - B. 2 triangles and 4 rectangles
 - C. 6 rectangles
 - D. 2 triangles and 3 rectangles
8. A solid with 5 faces, 9 edges and 6 vertices is a
 - A. triangular prism
 - B. square-based pyramid
 - C. cone
 - D. cuboid
9. When a cuboid of length l , width w and height h is unfolded, the six rectangles have a total area of
 - A. $l \times w \times h$
 - B. $lw + lh + wh$
 - C. $2(lw + lh + wh)$
 - D. $4(lw + lh + wh)$
10. The lateral (side) surface area of any prism equals
 - A. area of cross-section $\times$ length
 - B. perimeter of cross-section $\times$ length
 - C. $2 \times$ area of cross-section $\times$ length
 - D. half the perimeter $\times$ length
11. The total length of all the edges of a cuboid l , w , h is
 - A. $l + w + h$
 - B. $6(l + w + h)$
 - C. $4(l + w + h)$
 - D. $2(l + w + h)$
12. A prism whose base is an n -sided polygon has how many rectangular lateral faces?
 - A. $n - 2$
 - B. n
 - C. $n + 2$
 - D. $2n$

- 13.** The surface area of a cuboid $4\text{ cm} \times 3\text{ cm} \times 2\text{ cm}$ is
A. 52 cm^2
B. 24 cm^2
C. 40 cm^2
D. 94 cm^2
- 15.** The surface area of a cube is 150 cm^2 . Its side is
A. 10 cm
B. 25 cm
C. 5 cm
D. 6 cm
- 17.** The net folds into a cuboid. Its surface area is
A. 108 cm^2
B. 72 cm^2
C. 96 cm^2
D. 54 cm^2
- 19.** A cuboid has length 6 cm , width 4 cm and surface area 148 cm^2 . Its height is
A. 4 cm
B. 5 cm
C. 6 cm
D. 7 cm
- 21.** When the side of a cube is doubled, its surface area becomes
A. the same
B. 2 times as large
C. 4 times as large
D. 8 times as large
- 23.** The four walls of a room $5\text{ m} \times 4\text{ m}$ and height 3 m are to be painted. The painted area is
A. 60 m^2
B. 94 m^2
C. 54 m^2
D. 47 m^2
- 25.** A prism has triangular cross-section of area A and perimeter P , with length l . Its surface area is
A. $A + Pl$
B. $2A + \frac{1}{2}Pl$
C. $2A + Pl$
D. $2(A + Pl)$
- 14.** The surface area of a cube of side 4 cm is
A. 48 cm^2
B. 64 cm^2
C. 16 cm^2
D. 96 cm^2
- 16.** The surface area of a cuboid $5\text{ cm} \times 4\text{ cm} \times 3\text{ cm}$ is
A. 60 cm^2
B. 84 cm^2
C. 120 cm^2
D. 94 cm^2
- 18.** An open box with no lid has base $6\text{ cm} \times 4\text{ cm}$ and height 5 cm . The total area of its five faces is
A. 120 cm^2
B. 148 cm^2
C. 100 cm^2
D. 124 cm^2
- 20.** The surface area of a cuboid $10\text{ cm} \times 8\text{ cm} \times 5\text{ cm}$ is
A. 340 cm^2
B. 230 cm^2
C. 400 cm^2
D. 170 cm^2
- 22.** A gift box measures $20\text{ cm} \times 15\text{ cm} \times 10\text{ cm}$. Ignoring overlaps, the wrapping paper needed to cover it completely is
A. 700 cm^2
B. $1\,300\text{ cm}^2$
C. 650 cm^2
D. $3\,000\text{ cm}^2$
- 24.** Two cubes of side 2 cm are joined face to face to form a cuboid. The surface area of the cuboid is
A. 32 cm^2
B. 40 cm^2
C. 48 cm^2
D. 44 cm^2
- 26.** A triangular prism has right-triangle cross-section with sides 3 cm , 4 cm and 5 cm , and length 10 cm . Its surface area is
A. 120 cm^2
B. 144 cm^2
C. 96 cm^2
D. 132 cm^2

- 27.** The net folds into a triangular prism. Its surface area is
- A. 120 cm^2
 - B. 108 cm^2
 - C. 84 cm^2
 - D. 96 cm^2
- 29.** A triangular prism has right-triangle cross-section 6 cm–8 cm–10 cm and length 20 cm. Its surface area is
- A. 480 cm^2
 - B. 528 cm^2
 - C. 576 cm^2
 - D. 432 cm^2
- 31.** A net of 2 triangles and 3 rectangles folds into a
- A. cuboid
 - B. pyramid
 - C. cone
 - D. triangular prism
- 33.** A prism has triangular cross-section of area 24 cm^2 and perimeter 24 cm. Its length is 10 cm. Its surface area is
- A. 240 cm^2
 - B. 264 cm^2
 - C. 288 cm^2
 - D. 312 cm^2
- 35.** The number of faces of a triangular prism is
- A. 6
 - B. 7
 - C. 5
 - D. 4
- 37.** The difference between the surface areas of a cube of side 4 cm and a cuboid $6 \text{ cm} \times 4 \text{ cm} \times 2 \text{ cm}$ is
- A. 8 cm^2
 - B. 4 cm^2
 - C. 16 cm^2
 - D. 24 cm^2
- 39.** A cuboid has a square base of side 3 cm and height 8 cm. Its surface area is
- A. 132 cm^2
 - B. 96 cm^2
 - C. 108 cm^2
 - D. 114 cm^2
- 28.** A prism has isosceles-triangle cross-section with base 8 cm, height 3 cm and equal sides 5 cm. Its length is 12 cm. Its surface area is
- A. 240 cm^2
 - B. 216 cm^2
 - C. 228 cm^2
 - D. 264 cm^2
- 30.** The tent is a triangular prism with isosceles triangular ends (base 2 m, height 0.75 m, slant sides 1.25 m) and length 3 m. The cloth for the two ends and the two slanted faces (no floor) is
- A. 9 m^2
 - B. 10.5 m^2
 - C. 7.5 m^2
 - D. 8.25 m^2
- 32.** A triangular prism has equilateral-triangle cross-section of side 6 cm (triangle height $\approx 5.2 \text{ cm}$) and length 10 cm. Its surface area is about
- A. 195.6 cm^2
 - B. 211.2 cm^2
 - C. 226.8 cm^2
 - D. 180 cm^2
- 34.** In the net of a triangular prism, the two congruent triangles become the
- A. lateral faces of the prism
 - B. diagonals of the prism
 - C. vertices of the prism
 - D. bases (ends) of the prism
- 36.** Surface area of a solid is measured in
- A. units
 - B. cubic units
 - C. units of length
 - D. square units
- 38.** The net of a prism with a pentagonal base contains how many rectangles?
- A. 6
 - B. 5
 - C. 7
 - D. 3

- 40.** The total edge length of a cube is 60 cm. Its surface area is
- A. 150 cm^2
 - B. 100 cm^2
 - C. 125 cm^2
 - D. 300 cm^2

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Identifying solids and drawing nets

Question 2

(a) Deriving the cuboid surface-area formula

Question 3

(a) Cuboid surface-area problems

Question 4

(a) Deriving the triangular-prism surface area

Question 5

(a) Triangular-prism problems

Question 6

(a) Applications

Chapter 5: Ch14

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. With O as the origin, the position vector of M is
 - A. (3, 2)
 - B. (-2, -3)
 - C. (2, -3)
 - D. (2, 3)
2. With O as the origin, the position vector of N is
 - A. (2, -1)
 - B. (-1, 2)
 - C. (-2, 1)
 - D. (1, -2)
3. A movement of 3 units right and 2 units down is written as the column vector
 - A. (3, 2)
 - B. (-3, 2)
 - C. (2, 3)
 - D. (3, -2)
4. The magnitude of the vector (3, 4) is
 - A. 7
 - B. 5
 - C. 12
 - D. 25
5. The vector that translates A(1, 2) to B(4, 6) is
 - A. (3, 4)
 - B. (-3, -4)
 - C. (4, 3)
 - D. (5, 8)
6. The negative of the vector (-3, 4) is
 - A. (3, 4)
 - B. (-3, -4)
 - C. (3, -4)
 - D. (4, -3)
7. Two vectors are equal when they have
 - A. the same magnitude and the same direction
 - B. the same magnitude only
 - C. the same direction only
 - D. the same starting point
8. $2 \times (2, -3) =$
 - A. (2, -6)
 - B. (4, 6)
 - C. (4, -3)
 - D. (4, -6)
9. Two vectors are parallel when
 - A. their dot product is zero
 - B. their magnitudes are equal
 - C. one is a scalar multiple of the other
 - D. they start from the same point
10. The vector (6, 8) is parallel to
 - A. (4, 3)
 - B. (3, 4)
 - C. (-3, 4)
 - D. (8, 6)
11. If (k, 6) is parallel to (2, 3), then k =
 - A. 6
 - B. 9
 - C. 4
 - D. 2
12. The vectors (3x, 2) and (6, x) are parallel when x is
 - A. 12
 - B. 2 or -2
 - C. 4
 - D. 6
13. Two vectors are perpendicular when

- A. their dot product is zero
- B. their magnitudes are equal
- C. one is a scalar multiple of the other
- D. they have the same direction

15. A vector perpendicular to $(2, 5)$ is

- A. $(5, -2)$
- B. $(5, 2)$
- C. $(-2, -5)$
- D. $(2, -5)$

14. A vector perpendicular to $(3, 4)$ is

- A. $(4, 3)$
- B. $(3, -4)$
- C. $(-4, 3)$
- D. $(-3, 4)$

- 17.** If $(x, 4)$ is perpendicular to $(2, -1)$, then $x =$
 A. 8
 B. 4
 C. 2
 D. -2
- 19.** The vectors $(2, -3)$ and $(-4, 6)$ are
 A. perpendicular
 B. parallel
 C. equal
 D. neither parallel nor perpendicular
- 21.** If $(x, 2)$ is parallel to $(3, 6)$, then $x =$
 A. 1
 B. 6
 C. 2
 D. 3
- 23.** The triangle law of vector addition states that $AB + BC =$
 A. AC
 B. CA
 C. BA
 D. CB
- 25.** $(2, 5) - (1, 2) =$
 A. $(1, 3)$
 B. $(3, 7)$
 C. $(1, -3)$
 D. $(3, 3)$
- 27.** The magnitude of PQ in question 26 is
 A. 100
 B. 14
 C. 8
 D. 10
- 29.** In the parallelogram law, the resultant of u and v is the vector along
 A. the shorter side
 B. the diagonal from the common start point
 C. either dashed side
 D. the longer of u and v
- 31.** If $a = (2, 3)$ and $b = (1, -1)$, then $2a + b =$
 A. $(3, 5)$
 B. $(5, 2)$
 C. $(4, 4)$
 D. $(5, 5)$
- 16.** The dot product of (a, b) and (c, d) is
 A. $ad + bc$
 B. $ab + cd$
 C. $ac - bd$
 D. $ac + bd$
- 18.** If $(p, 3)$ is parallel to $(4, 12)$, then $p =$
 A. 9
 B. 3
 C. 4
 D. 1
- 20.** The vectors $(1, 2)$ and $(2, 1)$ are
 A. perpendicular
 B. equal
 C. neither parallel nor perpendicular
 D. parallel
- 22.** A vector perpendicular to $(-3, 4)$ is
 A. $(3, 4)$
 B. $(-4, -3)$
 C. $(4, 3)$
 D. $(4, -3)$
- 24.** Using the grid, $a + b =$
 A. $(1, -5)$
 B. $(5, -1)$
 C. $(5, 1)$
 D. $(-5, 1)$
- 26.** If $P(1, -3)$ and $Q(7, 5)$, the column vector PQ is
 A. $(-6, -8)$
 B. $(6, 8)$
 C. $(6, -8)$
 D. $(8, 6)$
- 28.** Triangle PQR with $P(1, -3)$, $Q(7, 5)$, $R(-3, 5)$ is isosceles because
 A. $PQ = QR = 10$ units
 B. $PQ = PR$
 C. $QR = PR$
 D. all sides are equal
- 30.** The resultant of $(2, 1)$ and $(3, 4)$ is
 A. $(5, 5)$
 B. $(5, 4)$
 C. $(1, 3)$
 D. $(6, 4)$

- 33.** In parallelogram PQRS, the vector PQ is equal to
A. SR
B. QS
C. RS
D. QR
- 35.** A boat sails (3, 4) and then (-1, 2). Its total displacement is
A. (2, 2)
B. (-4, -6)
C. (4, 2)
D. (2, 6)
- 37.** If $\overrightarrow{AB} = (2, 5)$ and A is (1, 1), then B is
A. (1, -4)
B. (-1, -4)
C. (6, 3)
D. (3, 6)
- 39.** $(2, 3) + (-2, -3)$ equals
A. (-4, -6)
B. (4, 6)
C. (0, 6)
D. the zero vector (0, 0)
- 32.** If $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (2, 1)$, then $3\mathbf{a} - 2\mathbf{b} =$
A. (-1, -4)
B. (7, 4)
C. (-1, 4)
D. (1, 4)
- 34.** PQRS is a parallelogram with P(x, y), Q(5, 7), R(2, 4) and S(1, 3). The values of x and y are
A. $x = 6, y = 4$
B. $x = 3, y = 5$
C. $x = 4, y = 6$
D. $x = 5, y = 6$
- 36.** The magnitude of the resultant of (3, 0) and (0, 4) is
A. 25
B. 5
C. 7
D. 12
- 38.** If M is the midpoint of PQ, then PM equals
A. PQ
B. $-\frac{1}{2}$ PQ
C. $\frac{1}{2}$ PQ
D. 2 PQ
- 40.** A displacement of 5 km due east of O, written as a column vector, is
A. (0, 5)
B. (5, 0)
C. (-5, 0)
D. (5, 5)

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Position vectors and notation

Question 2

- (a) Parallel vectors

Question 3

- (a) Perpendicular vectors

Question 4

- (a) The triangle law of addition

Question 5

- (a) The parallelogram law of addition

Question 6

- (a) Vector journeys

Chapter 6: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. An enlargement is completely defined by
 - A. a centre and a scale factor
 - B. an angle and a distance
 - C. a line of symmetry
 - D. a direction and a length
2. Under an enlargement, the angles of the object are
 - A. halved
 - B. multiplied by the scale factor
 - C. unchanged
 - D. doubled
3. Under an enlargement of scale factor k , the lengths of the object are
 - A. unchanged
 - B. multiplied by k^2
 - C. increased by k
 - D. multiplied by $|k|$
4. Under an enlargement of scale factor k , the area of the object is multiplied by
 - A. k^2
 - B. k
 - C. $2k$
 - D. $\sqrt{k}$
5. An enlargement with scale factor 1 gives an image that is
 - A. smaller than the object
 - B. a mirror image of the object
 - C. twice the object
 - D. congruent to the object
6. An enlargement with scale factor -1 gives an image that is
 - A. a smaller copy of the object
 - B. a larger copy of the object
 - C. the object turned through 180° about the centre
 - D. a translation of the object
7. The lines joining corresponding vertices of an object and its image under an enlargement all pass through
 - A. the origin
 - B. the midpoint of the object
 - C. the image
 - D. the centre of enlargement
8. An enlargement with scale factor 2 changes the perimeter of the object by a factor of
 - A. 8
 - B. 2
 - C. $\sqrt{2}$
 - D. 4
9. Which situation is an example of enlargement?
 - A. flipping a page of a book
 - B. turning a door on its hinges
 - C. printing a small photograph as a poster
 - D. sliding a box across the floor
10. A projector throws an image on a screen. The image is an enlargement because it is
 - A. the same size but a different shape
 - B. the same shape but a different size
 - C. a reflection of the slide
 - D. a translation of the slide
11. The shadow of a triangle on a wall has the same shape but a larger size. This is an example of
 - A. enlargement

- B. rotation
- C. translation
- D. reflection

- 13.** Photocopying a document at 150% corresponds to a scale factor of
- A. 1.5
 - B. 0.15
 - C. 15
 - D. 150
- 15.** An object length of 4 cm has an image length of 12 cm. The scale factor is
- A. 3
 - B. $\frac{1}{3}$
 - C. 8
 - D. 48
- 17.** The image lies on the opposite side of the centre and is twice as large as the object. The scale factor is
- A. $-\frac{1}{2}$
 - B. $\frac{1}{2}$
 - C. -2
 - D. 2
- 19.** The scale factor of the enlargement that maps P onto Q is
- A. 2
 - B. $-\frac{1}{2}$
 - C. $\frac{1}{2}$
 - D. -2
- 21.** The image of (2, 3) under an enlargement with centre O and scale factor 3 is
- A. (9, 6)
 - B. (5, 6)
 - C. (2, 9)
 - D. (6, 9)
- 23.** An enlargement with centre (1, 1) and scale factor 2 maps (3, 1) to
- A. (7, 1)
 - B. (4, 2)
 - C. (5, 1)
 - D. (6, 2)
- 12.** Which of the following is NOT an enlargement?
- A. blowing up a passport photo to poster size
 - B. a reflection in a plane mirror
 - C. zooming in on a phone screen
 - D. the shadow of a chair at sunset
- 14.** Zooming $\times 3$ on a phone map corresponds to a scale factor of
- A. 0.3
 - B. 30
 - C. $\frac{1}{3}$
 - D. 3
- 16.** An object length of 6 cm has an image length of 3 cm. The scale factor is
- A. 2
 - B. $\frac{1}{2}$
 - C. 3
 - D. 18
- 18.** The scale factor of the enlargement that maps P onto Q is
- A. 4
 - B. 2
 - C. $\frac{1}{2}$
 - D. -2
- 20.** The scale factor of the enlargement that maps P onto Q is
- A. -2
 - B. $\frac{1}{2}$
 - C. $-\frac{1}{2}$
 - D. 2
- 22.** The image of (4, -2) under an enlargement with centre O and scale factor -2 is
- A. $(-8, 4)$
 - B. $(8, -4)$
 - C. $(-8, -4)$
 - D. $(2, -4)$
- 24.** When $0 < k < 1$, the image is
- A. larger and on the opposite side of the centre
 - B. larger and on the same side of the centre
 - C. smaller and on the opposite side of the centre
 - D. smaller and on the same side of

the centre

- 25.** When $k < -1$, the image is
- A. larger and on the same side of the centre
 - B. smaller and on the opposite side of the centre
 - C. larger and on the opposite side of the centre
 - D. smaller and on the same side of

the centre

- 27.** The ratio of the perimeter of the image to the perimeter of the object equals
- $1/|k|$
 - k^2
 - $2k$
 - $|k|$
- 29.** An image has area 4 cm^2 and the object has area 36 cm^2 . A possible scale factor is
- 9
 - $1/3$ or $-1/3$
 - 3 or -3
 - $1/9$
- 31.** Congruent shapes have
- the same shape but different sizes
 - the same shape and the same size
 - different shapes but the same size
 - different shapes and sizes
- 33.** The image under any enlargement is always
- similar to the object
 - a mirror image of the object
 - congruent to the object
 - larger than the object
- 35.** Similar triangles with sides in the ratio $2 : 3$ have areas in the ratio
- $8 : 27$
 - $4 : 9$
 - $\sqrt{2} : \sqrt{3}$
 - $2 : 3$
- 37.** Which pair of shapes is ALWAYS similar?
- any two rectangles
 - any two isosceles triangles
 - any two rhombuses
 - any two squares
- 39.** Similar shapes with scale factor 4 have areas in the ratio
- $1 : 64$
 - $1 : 4$
 - $1 : 16$
 - $1 : 8$
- 26.** Under an enlargement with centre O and scale factor k , the point (x, y) maps to
- (kx, y)
 - (kx, ky)
 - $(x/k, y/k)$
 - $(x + k, y + k)$
- 28.** An object has area 10 cm^2 . With scale factor 3, the image area is
- 90 cm^2
 - 30 cm^2
 - 13 cm^2
 - 900 cm^2
- 30.** The centre of the enlargement that maps P onto Q is
- $(1, 0)$
 - $(1, 1)$
 - $(0, 0)$
 - $(2, 2)$
- 32.** Similar shapes have
- equal areas
 - equal perimeters
 - equal angles and proportional sides
 - equal sides only
- 34.** The image of an enlargement is congruent to the object when
- $k = 2$
 - $k = 0$
 - $|k| = 1$
 - $k > 1$
- 36.** Two similar rectangles have shorter sides 3 cm and 9 cm. If the longer side of the smaller is 5 cm, the longer side of the larger is
- 15 cm
 - 45 cm
 - 11 cm
 - 25 cm
- 38.** Which pair of shapes is NOT necessarily similar?
- any two rectangles
 - any two equilateral triangles
 - any two squares
 - any two circles

- 40.** An object with sides 2 cm and 3 cm maps to a shape with sides 6 cm and 9 cm. The shapes are
- A. congruent
 - B. similar with scale factor 9
 - C. neither similar nor congruent
 - D. similar with scale factor 3

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Everyday enlargements and properties

Question 2

(a) Drawing enlargements on a grid

Question 3

(a) Scale factors from diagrams

Question 4

(a) Enlargement coordinates and areas

Question 5

(a) Congruent and similar shapes

Question 6

(a) Verifying similarity

Chapter 7: Ch16

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Quantitative data are
 - A. descriptive data expressed in words
 - B. data taken from existing records
 - C. numerical data obtained by counting or measuring
 - D. data collected by interview only
2. Qualitative data are
 - A. descriptive (non-numerical) data
 - B. grouped data
 - C. secondary data
 - D. numerical data
3. Musa wants to find the books that are most popular among his friends. The best method of collecting the data is
 - A. using a database
 - B. an experiment
 - C. a questionnaire or survey
 - D. observation
4. To find the most common mode of travel by learners in a school, the best method is
 - A. an experiment
 - B. observation only
 - C. measurement with instruments
 - D. a survey using a questionnaire
5. Counting vehicles passing a junction between 7 am and 8 am is best done by
 - A. observation
 - B. questionnaire
 - C. experiment
 - D. interview
6. Life-testing thirty bulbs to the nearest hour is an example of data collection by
 - A. using existing records
 - B. survey
 - C. interview
 - D. experiment
7. Asking respondents questions face to face and recording their answers is called
 - A. an interview
 - B. a questionnaire
 - C. observation
 - D. an experiment
8. Using the school's attendance register, kept over the years, to study attendance is using
 - A. observation
 - B. existing records (secondary data)
 - C. an interview
 - D. primary data from an experiment
9. The correct tally for 13 is
 - A. thirteen separate single strokes as standard practice
 - B. one group of five and three strokes
 - C. three complete groups of five
 - D. two complete tally groups of five and three extra strokes
10. The sum of the frequencies in a frequency table always equals
 - A. the number of data values
 - B. the range
 - C. the mode
 - D. the number of classes
11. The mode of the data 2, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 7, 7 is
 - A. 6

- B. 7
- C. 4
- D. 5

13. The range of 12, 5, 9, 3, 8 is

- A. 7
- B. 12
- C. 9
- D. 8

12. The mode of 4, 7, 7, 9, 7, 4 is

- A. 6
- B. 7
- C. 4
- D. 9

- 15.** The value represented by $2 \mid 4$ is
A. 240
B. 24
C. 42
D. 6
- 17.** In a bar chart, the bars
A. are always drawn horizontal
B. must show continuous data
C. touch each other
D. are separated by gaps
- 19.** In a pie chart, a frequency of 3 out of a total of 30 gives a sector angle of
A. 30°
B. 36°
C. 60°
D. 108°
- 21.** In a 10×10 waffle diagram, each small square represents
A. 5%
B. 1%
C. 10%
D. 0.1%
- 23.** The class width of the class interval 168–171 (whole-number data) is
A. 5
B. 4
C. 8
D. 3
- 25.** In the data 2, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 7, 7 the frequency of 4 is
A. 3
B. 5
C. 6
D. 4
- 27.** In a histogram the bars touch each other because
A. the data are continuous
B. the bars must be equal in height
C. the data are categorical
D. all frequencies are equal
- 14.** The median of 3, 5, 7, 9 is
A. 7
B. 8
C. 5
D. 6
- 16.** The number of data values shown is
A. 9
B. 7
C. 8
D. 10
- 18.** The representation best suited to showing change over time is a
A. line graph
B. waffle diagram
C. pie chart
D. pictograph
- 20.** In a pictograph one symbol represents 10 books. Four and a half symbols represent
A. 50 books
B. 14 books
C. 45 books
D. 40 books
- 22.** A box-and-whisker plot displays the median together with
A. the mode only
B. the frequency of every value
C. the mean only
D. the quartiles and extreme values
- 24.** A grouped table has frequencies 5, 9, 12 and 7. The modal class is the class with frequency
A. 7
B. 5
C. 12
D. 9
- 26.** The diagram used to represent continuous grouped data is a
A. pie chart
B. waffle diagram
C. histogram
D. pictograph
- 28.** A frequency table has frequencies 3, x, 6, 5, 2 with a total of 20. The value of x is
A. 3
B. 6

- C. 4
- D. 8

30. The modal class is

- A. 30-39
- B. 50-59
- C. 10-19
- D. 20-29

29. In a histogram, the mode is read from

- A. the last bar
- B. the middle bar
- C. the shortest bar
- D. the tallest bar

31. The total number of students is

- A. 25

- B. 30
C. 35
D. 10
- 33.** The difference between the greatest and the least frequency is
A. 10
B. 7
C. 3
D. 4
- 35.** For grouped data, the mode is given as
A. the modal class
B. the midpoint of the first class
C. half of the greatest frequency
D. a single exact value
- 37.** If the mode of the data 2, 4, 4, x, 6 is 4, then x could be
A. 2
B. 4
C. 6
D. any number
- 39.** Five classes have frequencies 2, 5, 8, 5, 2. The modal class is
A. the fifth class
B. the third class
C. the first class
D. the second class
- 32.** The frequency of the class 20–29 is
A. 6
B. 10
C. 4
D. 7
- 34.** A histogram has no gaps between its bars because the data are
A. categorical
B. discrete categories
C. continuous (grouped measurements)
D. qualitative
- 36.** Bulb lifespans (hours): 164–167 $f = 4$; 168–171 $f = 7$; 172–175 $f = 11$; 176–179 $f = 8$. The modal class is
A. 176–179
B. 164–167
C. 172–175
D. 168–171
- 38.** The class that contains the median mark is
A. 30–39
B. 20–29
C. 40–49
D. 10–19
- 40.** A histogram has total frequency 40. Four bars show 8, 12, 10 and 6. The fifth bar shows
A. 6
B. 10
C. 4
D. 14

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Data types and collection methods

Question 2

(a) Ungrouped frequency table

Question 3

(a) Grouped frequency table

Question 4

(a) Stem-and-leaf, pictograph and pie chart

Question 5

(a) Histograms

Question 6

(a) Create and solve

Chapter 8: Ch17

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A survey question 'Overall, don't you think the teaching of mathematics is very good?' is biased because it
 - A. is too short
 - B. uses difficult words
 - C. leads respondents towards answering 'yes'
 - D. has too many options
3. Ama surveyed only the school's football team about sport-related injuries. The main problem is that
 - A. the sample is only footballers (selection bias)
 - B. injuries never happen
 - C. football is not a sport
 - D. the team is too small to exist
5. According to the curriculum, which group of factors may introduce bias into data collection?
 - A. mean, median and mode
 - B. language, ethics, cost, time and timing, privacy, cultural sensitivity
 - C. bar charts and pie charts
 - D. rulers, compasses and protractors
7. A survey on market sales conducted only between 7 am and 9 am is likely to be biased because of
 - A. timing — morning traders are not representative of the whole day
 - B. the cost of printing
 - C. the language used
 - D. the privacy of traders
9. The mean of 4, 6, 8, 10, 12 is
 - A. 40
 - B. 8
 - C. 6
 - D. 10
2. A better, neutral version of the question is 'Overall, how will you rate the teaching of mathematics?' with the scale
 - A. agree - disagree only
 - B. very poor - poor - fair - good - very good
 - C. true - false
 - D. yes - no
4. To remove the bias, Ama should ask
 - A. more footballers only
 - B. only the coaches
 - C. people in a variety of sports, contact and non-contact
 - D. only spectators
6. Asking 'If you eat pork, name your favourite method of cooking it' in a community where pork is forbidden may offend respondents. The influencing factor is
 - A. privacy
 - B. cost
 - C. cultural sensitivity
 - D. timing
8. Asking respondents to state their exact personal income in a public place raises the issue of
 - A. timing
 - B. sample size
 - C. cost
 - D. privacy
10. The mean of 5, 7, 9 is
 - A. 6
 - B. 8
 - C. 21
 - D. 7

- 11.** The median of 3, 8, 5, 2, 9 is
A. 8
B. 27
C. 5
D. 3
- 13.** The range of 14, 3, 9, 20, 7 is
A. 11
B. 20
C. 53
D. 17
- 15.** The range of the bulb lifespans 164–179 hours is
A. 343 hours
B. 14 hours
C. 15 hours
D. 16 hours
- 17.** The sum of n values equals
A. $\text{mean} \div n$
B. $\text{mean} \times n$
C. $n \div \text{mean}$
D. $\text{mean} + n$
- 19.** The median of 4, 6, 8, 10 is
A. 7
B. 6
C. 8
D. 28
- 21.** When a data set contains extreme values, the most reliable measure of central tendency is usually the
A. median
B. sum
C. mean
D. range
- 23.** The mean of the first five natural numbers 1, 2, 3, 4, 5 is
A. 3
B. 3.5
C. 15
D. 2.5
- 25.** A shopkeeper ordering bulbs to sell would mostly rely on which statistic, to stock the lifespan customers most often get?
A. the maximum
B. the minimum
C. the mode
D. the range
- 12.** The mode of 2, 3, 3, 4, 5, 3, 2 is
A. 5
B. 3
C. 2
D. 4
- 14.** Bulb lifespans were recorded to the nearest hour from 164 to 179. The minimum and maximum lifespans are
A. 164 hours and 179 hours
B. 164 hours and 176 hours
C. 168 hours and 179 hours
D. 167 hours and 176 hours
- 16.** Values 2, 3, 4 occur with frequencies 3, 5, 2. The mean is
A. 3.0
B. 2.5
C. 9.0
D. 2.9
- 18.** The mean of four numbers is 6. If three of them are 5, 7 and 4, the fourth is
A. 24
B. 6
C. 10
D. 8
- 20.** The mode of the data is
A. 3 siblings
B. 20 siblings
C. 2 siblings
D. 8 siblings
- 22.** The measure of central tendency that uses every value in the data set is the
A. median
B. mode
C. range
D. mean
- 24.** The mean age of 10 people is 15 years. A 26-year-old joins them. The new mean is
A. 41 years
B. 16 years
C. 15 years
D. 17 years

- 27.** The number of students represented is
- A. 40
 - B. 20
 - C. 8
 - D. 24
- 29.** The mean of 2, 3, 3, 5, 7 is 4. If the 7 is replaced by 300, the new mean is
- A. 20.0
 - B. 62.6
 - C. 313
 - D. 4.0
- 31.** The measure LEAST affected by adding one extreme value to 2, 3, 3, 5, 7 is the
- A. mean
 - B. sum
 - C. median (and the mode)
 - D. range
- 33.** Adding one extreme value that occurs only once to a data set leaves which measure unchanged?
- A. the range
 - B. the mean
 - C. the mode
 - D. the sum
- 35.** The mean of five values is 8. A sixth value of 20 is added. The new mean is
- A. 8
 - B. 28
 - C. 10
 - D. 14
- 37.** A data set has mean 9, median 9 and mode 9. One extreme value 100 is added. Which measure stays 9?
- A. the mean
 - B. the mode
 - C. the sum
 - D. all of them
- 39.** Five workers earn GH¢ 900–1 050 but the manager earns GH¢ 10 000. The typical salary is best described by the
- A. sum
 - B. mean
 - C. maximum
 - D. median
- 26.** The median lifespan of the bulbs tells the shopkeeper that
- A. half of the bulbs lasted at least that long
 - B. most bulbs failed before that time
 - C. the lifespans are all equal
 - D. all bulbs lasted that long
- 28.** The mean number of siblings is
- A. 2
 - B. 1
 - C. 3
 - D. 4
- 30.** The measure of central tendency most affected by an extreme value is the
- A. frequency
 - B. median
 - C. mode
 - D. mean
- 32.** The median of 3, 4, 5, 6, 7 is 5. If 7 is replaced by 100, the new median is
- A. 25
 - B. 6
 - C. 100
 - D. 5
- 34.** Data 10, 12, 12, 14, 16: replacing 16 with 160 changes which measure the most?
- A. the mean
 - B. the mode
 - C. the median
 - D. none of them
- 36.** Bulbs with mean lifespan 170 hours: one bulb is replaced by a 300-hour bulb. The mean will
- A. become 300
 - B. decrease
 - C. stay exactly 170
 - D. increase
- 38.** A value that is far away from the rest of the data is called
- A. an outlier (extreme value)
 - B. a class interval
 - C. a frequency
 - D. a mode

- 40.** Data 5, 6, 7, 8, 9 has mean 7. Adding 70 gives
- A. mean 7.5 and median 17.5
 - B. mean 17.5 and median 7.5
 - C. mean 12 and median 7
 - D. mean 7 and median 70

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Bias in data collection

Question 2

- (a) Descriptive statistics of ungrouped data

Question 3

- (a) Descriptive statistics of grouped data

Question 4

- (a) Working with the mean

Question 5

- (a) Effect of extreme data

Question 6

- (a) Interpreting with extreme values

Chapter 9: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The sample space for tossing a coin once is
 - A. {Heads, Tails}
 - B. {1, 2}
 - C. {Heads}
 - D. {0, 1, 2}
2. The sample space for rolling a fair die has
 - A. 4 outcomes
 - B. 36 outcomes
 - C. 6 equally likely outcomes
 - D. 2 outcomes
3. The number of outcomes in the sample space when two coins are tossed is
 - A. 16
 - B. 4
 - C. 2
 - D. 8
4. The probability of any event lies between
 - A. 0 and 1
 - B. -1 and 1
 - C. 0 and 100
 - D. 1 and 10
5. The probability of a certain event and of an impossible event are respectively
 - A. 1 and 1
 - B. 1 and 0
 - C. 0 and 0
 - D. 0 and 1
6. For any event E, $P(E) + P(\text{not } E) =$
 - A. 1
 - B. $\frac{1}{2}$
 - C. 0
 - D. 2
7. The probability of an event is calculated as
 - A. $\text{total} \div \text{favourable}$
 - B. $\text{favourable outcomes} \div \text{total outcomes}$
 - C. $\text{favourable} - \text{total}$
 - D. $\text{unfavourable} \div \text{favourable}$
8. The fraction $\frac{1}{4}$ expressed as a decimal and as a percentage is
 - A. 0.75 and 75%
 - B. 2.5 and 2.5%
 - C. 0.25 and 25%
 - D. 0.4 and 40%
9. Drawing two bottle tops from a bag WITHOUT replacement gives
 - A. dependent events
 - B. certain events
 - C. independent events
 - D. impossible events
10. A bag has 3 red, 2 green and 1 pink bottle top. The probability that the first draw is red is
 - A. $\frac{1}{3}$
 - B. $\frac{1}{6}$
 - C. $\frac{2}{3}$
 - D. $\frac{1}{2}$
11. If the first draw (without replacement) is red, the probability that the second draw is red is
 - A. $\frac{2}{5}$

- B. $\frac{3}{5}$
- C. $\frac{1}{6}$
- D. $\frac{3}{6}$

- 13.** Drawing two tops (without replacement) from 3 red, 2 green, 1 pink, the number of possible ordered colour pairs is
- A. 6
 - B. 8 (PP is impossible)
 - C. 9
 - D. 36

- 12.** In the bottle-top experiment, does the first trial affect the second?
- A. Only when drawing with replacement
 - B. No — the bag stays the same
 - C. Only if a pink top is drawn
 - D. Yes — the first draw changes what is left in the bag

- 15.** A bag has 2 blue and 3 yellow marbles. Two are drawn without replacement. $P(\text{blue then blue}) =$
A. $4/25$
B. $1/10$
C. $2/10$
D. $1/4$
- 17.** The probability that the first card drawn from a full deck is an Ace is
A. $1/13$
B. $4/13$
C. $1/52$
D. $1/4$
- 19.** The probability of drawing an Ace and then a Jack, without replacement, is
A. $16/2652 \times 2$
B. $1/221$
C. $4/663$
D. $8/52$
- 21.** Drawing without replacement changes the probability of the second draw because the
A. outcomes become equally likely
B. number of colours increases
C. total number of items left decreases
D. bag becomes heavier
- 23.** With the same bag, $P(\text{both sweets the same colour}) =$
A. $1/2$
B. $4/7$
C. $3/7$
D. $3/14$
- 25.** $3/4$ expressed as a percentage is
A. 34%
B. 43%
C. 75%
D. 80%
- 27.** $1/8$ expressed as a decimal is
A. 1.8
B. 0.8
C. 0.18
D. 0.125
- 14.** Drawing WITH replacement gives
A. mutually exclusive events
B. impossible events
C. dependent events
D. independent events
- 16.** With the same bag, $P(\text{yellow then blue}) =$
A. $3/20$
B. $6/25$
C. $3/10$
D. $1/10$
- 18.** The probability of drawing two Aces in two trials without replacement is
A. $12/52$
B. $1/169$
C. $1/17$
D. $1/221$
- 20.** After one Ace has been removed from a full deck, the probability the next card is an Ace is
A. $3/52$
B. $4/51 \times 4$
C. $1/13$
D. $1/17$
- 22.** A bag has 4 red and 4 blue sweets. Two are taken without replacement. $P(\text{red then red}) =$
A. $1/4$
B. $3/14$
C. $12/64$
D. $16/56$
- 24.** For two dependent events A then B, $P(A \text{ and } B) =$
A. $P(A) - P(B)$
B. $P(A) \times P(B \text{ given } A)$
C. $P(A) \times P(B)$ always
D. $P(A) + P(B)$
- 26.** 0.2 expressed as a fraction in lowest terms is
A. $1/5$
B. $1/2$
C. $2/100$
D. $2/5$

- 29.** A probability of $\frac{3}{5}$ means the ratio of favourable to unfavourable outcomes is
- A. 2 : 3
 - B. 5 : 3
 - C. 3 : 2
 - D. 3 : 5
- 31.** The sum of the three first-stage branch probabilities ($\frac{3}{6} + \frac{2}{6} + \frac{1}{6}$) is
- A. 1
 - B. 0
 - C. 6
 - D. $\frac{1}{2}$
- 33.** $\frac{1}{13}$ expressed as a percentage to 1 decimal place is about
- A. 0.77%
 - B. 7.7%
 - C. 7.5%
 - D. 13%
- 35.** From the 3-red, 2-green, 1-pink bag, two tops drawn without replacement: $P(\text{green then green}) =$
- A. $\frac{1}{9}$
 - B. $\frac{4}{36}$
 - C. $\frac{2}{15}$
 - D. $\frac{1}{15}$
- 37.** $\frac{1}{10}$ expressed as a percentage is
- A. 1%
 - B. 0.1%
 - C. 90%
 - D. 10%
- 39.** $\frac{4}{663}$ as a percentage is about
- A. 0.6%
 - B. 60%
 - C. 6%
 - D. 0.06%
- 28.** 40% expressed as a fraction in lowest terms is
- A. $\frac{4}{5}$
 - B. $\frac{2}{5}$
 - C. $\frac{40}{10}$
 - D. $\frac{1}{40}$
- 30.** The probability of the path $R \rightarrow R$ is
- A. $\frac{6}{11}$
 - B. $\frac{3}{6}$
 - C. $\frac{2}{5}$
 - D. $\frac{1}{5}$
- 32.** $\frac{1}{221}$ expressed as a percentage to 2 significant figures is about
- A. 0.22%
 - B. 45%
 - C. 4.5%
 - D. 0.45%
- 34.** A probability of $\frac{2}{5}$ gives the ratio favourable : unfavourable =
- A. 2 : 3
 - B. 2 : 5
 - C. 3 : 2
 - D. 5 : 2
- 36.** From the same bag, $P(\text{at least one red in two draws}) =$
- A. $\frac{9}{10}$
 - B. $\frac{1}{5}$
 - C. $\frac{4}{5}$
 - D. $\frac{3}{5}$
- 38.** $\frac{3}{10}$ expressed as a decimal is
- A. 0.03
 - B. 0.3
 - C. 0.33
 - D. 3.0
- 40.** A probability of 0.45 expressed as a ratio favourable : unfavourable is
- A. 45 : 100
 - B. 9 : 20
 - C. 9 : 11
 - D. 11 : 9

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Sample space and dependence

Question 2

(a) Tree diagram

Question 3

(a) Card experiments (curriculum exemplars)

Question 4

(a) Converting probabilities

Question 5

(a) Dependent draws from 4 red and 4 blue

Question 6

(a) At least one red

Chapter 10: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The number 38.4972 rounded to 2 decimal places is
 - A. 38.49
 - B. 38.497
 - C. 38.40
 - D. 38.50
2. The number 506,723,412 correct to 4 significant figures is
 - A. 507,000,000
 - B. 506,700,000
 - C. 506,723,400
 - D. 506,720,000
3. The number of significant figures in 70.030 is
 - A. 2
 - B. 3
 - C. 5
 - D. 4
4. The number 2,684,913 correct to 3 significant figures is
 - A. 2,680,000
 - B. 2,690,000
 - C. 2,684,000
 - D. 2,700,000
5. The number 9.996 rounded to 2 decimal places is
 - A. 9.96
 - B. 9.996
 - C. 10.00
 - D. 9.99
6. The number 0.0596 correct to 2 significant figures is
 - A. 0.0596
 - B. 0.060
 - C. 0.059
 - D. 0.06
7. The number of significant figures in 0.004050 is
 - A. 4
 - B. 3
 - C. 6
 - D. 7
8. The number 74.996 rounded to 1 decimal place is
 - A. 74.99
 - B. 75.9
 - C. 74.9
 - D. 75.0
9. The numbers 0.501, 0.0501, 0.51 and 0.051 arranged in ascending order are
 - A. 0.501, 0.51, 0.051, 0.0501
 - B. 0.0501, 0.051, 0.501, 0.51
 - C. 0.051, 0.0501, 0.501, 0.51
 - D. 0.0501, 0.501, 0.051, 0.51
10. Which of the following numbers rounds to 6.0 correct to 1 decimal place?
 - A. 5.94
 - B. 6.06
 - C. 5.9
 - D. 5.96
11. The largest whole number which rounds to 45,000 to the nearest thousand is
 - A. 44,500
 - B. 45,999
 - C. 45,499
 - D. 45,500
12. The smallest number which rounds to 8.3 correct to 1 decimal place is
 - A. 8.25
 - B. 8.31
 - C. 8.26
 - D. 8.24

- 13.** A whole number rounds to 700 correct to 2 significant figures. The largest possible value of the number is
- A. 750
 - B. 704
 - C. 799
 - D. 749
- 15.** I am a 5-digit number. My ten-thousands digit is 3. My thousands digit is twice my ten-thousands digit. My hundreds digit is 4 less than my thousands digit. My tens digit is 0 and my units digit is 9. The number is
- A. 36,209
 - B. 36,290
 - C. 36,609
 - D. 32,609
- 17.** The place value of the digit 8 in the number 4.0805 is
- A. $8/100$
 - B. $8/1,000$
 - C. 8
 - D. $8/10$
- 19.** The number 83,650,000 written in standard form is
- A. 8.365×10^6
 - B. 83.65×10^6
 - C. 8.365×10^8
 - D. 8.365×10^7
- 21.** The number 0.0000743 written in standard form is
- A. 7.43×10^{-6}
 - B. 7.43×10^{-4}
 - C. 7.43×10^{-5}
 - D. 74.3×10^{-6}
- 23.** Evaluate, leaving your answer in standard form: $(5.6 \times 10^4) + (3.2 \times 10^4)$
- A. 88×10^4
 - B. 8.8×10^4
 - C. 8.8×10^3
 - D. 8.8×10^8
- 25.** The population of a region is 33,475,870. Correct to 2 significant figures, this is
- 14.** By rounding each number to 1 significant figure, the estimated value of 62×19 is
- A. 600
 - B. 1,800
 - C. 1,200
 - D. 1,178
- 16.** The number in question 15 expressed in standard form is
- A. 3.6209×10^5
 - B. 3.6209×10^4
 - C. 36.209×10^3
 - D. 3.6209×10^3
- 18.** The value of the digit 5 in the number 2.0053 is
- A. 0.5
 - B. 0.05
 - C. 0.005
 - D. 0.0005
- 20.** Written as an ordinary number, 4.07×10^5 is
- A. 4,070,000
 - B. 407,000
 - C. 4,070
 - D. 40,700
- 22.** Written as a decimal, 2.05×10^{-3} is
- A. 0.00205
 - B. 0.0205
 - C. 0.000205
 - D. 0.205
- 24.** Evaluate, leaving your answer in standard form: $(6.4 \times 10^6) \div (8 \times 10^2)$
- A. 8×10^2
 - B. 8×10^4
 - C. 0.8×10^4
 - D. 8×10^3

- A. 33,500,000
- B. 34,000,000
- C. 33,000,000
- D. 33,400,000

27. Ama paid GH¢50.00 for items costing GH¢37.45. Her change, correct to 1 decimal place, is

- A. GH¢13.00
- B. GH¢12.60
- C. GH¢12.50
- D. GH¢12.55

26. A farmer harvested 127,483 kg of maize. Correct to 3 significant figures, the harvest is

- A. 127,000 kg
- B. 127,500 kg
- C. 120,000 kg
- D. 128,000 kg

- 29.** The number of significant figures in 0.03080 is
- A. 5
 - B. 2
 - C. 4
 - D. 3
- 31.** The number 504,673 rounded to the nearest ten thousand is
- A. 510,000
 - B. 500,000
 - C. 504,700
 - D. 505,000
- 33.** The distance between two towns is 8,345,000 m. Written in standard form, this is
- A. 8.345×10^7 m
 - B. 8.345×10^5 m
 - C. 8.345×10^6 m
 - D. 0.8345×10^7 m
- 35.** $\sqrt{63}$ lies between which two consecutive whole numbers?
- A. 7 and 8
 - B. 8 and 9
 - C. 31 and 32
 - D. 6 and 7
- 37.** Correct to 2 decimal places, $\sqrt{12}$ is
- A. 3.46
 - B. 3.45
 - C. 3.47
 - D. 6.00
- 39.** Which of the following numbers is a perfect square?
- A. 99
 - B. 20
 - C. 50
 - D. 81
- 28.** The length of a wire is 0.004567 m. Correct to 2 significant figures, the length is
- A. 0.0045 m
 - B. 0.005 m
 - C. 0.00457 m
 - D. 0.0046 m
- 30.** Written as an ordinary number, 3.678×10^4 is
- A. 36,780
 - B. 36.78
 - C. 367,800
 - D. 3,678
- 32.** Evaluate, leaving your answer in standard form: $(2.4 \times 10^3) \times (3 \times 10^2)$
- A. 72×10^5
 - B. 5.4×10^5
 - C. 7.2×10^6
 - D. 7.2×10^5
- 34.** The population of a city is 3.2×10^6 . Written in full, the population is
- A. 32,000,000
 - B. 3,200,000
 - C. 320,000
 - D. 3,020,000
- 36.** The best estimate of $\sqrt{75}$, correct to 1 decimal place, is
- A. 9.4
 - B. 7.5
 - C. 8.0
 - D. 8.7
- 38.** If $x^2 = 145$, the value of x correct to 1 decimal place is
- A. 11.9
 - B. 72.5
 - C. 12.0
 - D. 12.1
- 40.** Correct to 1 decimal place, $\sqrt{101}$ is
- A. 10.5
 - B. 10.0
 - C. 9.5
 - D. 10.1

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Rounding numbers.

Question 2

(a) Standard form.

Question 3

(a) Place value puzzle.

Question 4

(a) Upper and lower bounds.

Question 5

(a) Approximating square roots.

Question 6

(a) Integrated problems.

Chapter 11: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following numbers is irrational?
A. 0.75
B. $\frac{2}{3}$
C. $\sqrt{2}$
D. $\sqrt{49}$
2. The set of natural numbers is usually denoted by
A. Q
B. W
C. Z
D. N
3. Every integer is also a
A. rational number
B. whole number
C. irrational number
D. natural number
4. Which of the following is a whole number but NOT a natural number?
A. 1
B. 0
C. -1
D. $\frac{1}{2}$
5. The decimal expansion of $\frac{22}{7}$ is
A. non-terminating and non-repeating
B. irrational
C. terminating
D. non-terminating and repeating
6. $\sqrt{49}$ simplifies to 7, so it is
A. a prime number
B. a rational number
C. an irrational number
D. not a real number
7. Which of the following best describes the relationship between the set of natural numbers N and the set of integers Z?
A. $N \subset Z$
B. $Z \subset N$
C. $N = Z$
D. $N \cap Z =$
8. The intersection of the set of rational numbers and the set of irrational numbers is
A. the set of real numbers
B. the set of rational numbers
C. the empty set,
D. $\{0\}$
9. The union of the set of rational numbers and the set of irrational numbers gives
A. the set of whole numbers
B. the set of integers
C. the set of rational numbers
D. the set of real numbers
10. Which of the following is NOT a rational number?
A. 0.333...
B. $\sqrt{5}$
C. -8
D. $\frac{3}{4}$
11. π is best classified as
A. a whole number
B. an integer
C. an irrational number
D. a rational number
12. Which of the following numbers belongs to Z but NOT to W?
A. -6
B. 0.6
C. 0
D. 6

- 13.** If $A = \{\text{factors of } 12\}$ and $B = \{\text{factors of } 15\}$, then $A \cap B =$
- $\{1, 2, 3, 4, 6, 12\}$
 - $\{3, 5, 15\}$
 - $\{1, 3\}$
 - $\{1\}$
- 15.** The number of elements in $A \cup B$, for the sets in question 13, is
- 8
 - 2
 - 10
 - 6
- 17.** The set that contains all the elements under consideration is called the
- subset
 - complement
 - universal set
 - empty set
- 19.** A set which has no elements is called the
- universal set
 - unit set
 - subset
 - empty set
- 21.** In a class of 40 pupils, 25 like Mathematics and 20 like Science. If every pupil likes at least one of the two subjects, the number who like both subjects is
- 45
 - 15
 - 5
 - 20
- 23.** From the information in question 21, the number of pupils who like exactly one subject is
- 25
 - 15
 - 40
 - 35
- 25.** From the survey in question 24, the number of people who drink coffee only is
- 50
 - 20
 - 30
 - 40
- 14.** From the sets in question 13, $A \cup B =$
- $\{1, 2, 3, 4, 6, 12, 15\}$
 - $\{2, 4, 5, 6, 12, 15\}$
 - $\{1, 3\}$
 - $\{1, 2, 3, 4, 5, 6, 12, 15\}$
- 16.** In a Venn diagram, the overlapping region of two circles represents the
- union of the sets
 - intersection of the sets
 - complement of a set
 - universal set
- 18.** If $n(A) = 6$, $n(B) = 4$ and $n(A \cap B) = 2$, then $n(A \cup B) =$
- 24
 - 8
 - 10
 - 12
- 20.** If $P = \{2, 4, 6\}$ and $Q = \{4, 6, 8\}$, then $P \cap Q =$
- $\{4, 6\}$
 - $\{2, 4, 6, 8\}$
 - $\{2, 4, 6\}$
 - $\{2, 8\}$
- 22.** From the information in question 21, the number of pupils who like Mathematics only is
- 25
 - 20
 - 15
 - 5
- 24.** 100 people were surveyed: 60 drink tea, 50 drink coffee and 10 drink neither. The number who drink both tea and coffee is
- 20
 - 110
 - 10
 - 30
- 26.** From the survey in question 24, the probability that a person chosen at random drinks tea only is
- $1/2$
 - $2/5$
 - $3/10$
 - $3/5$

- 27.** In a village of 80 farmers, 50 grow beans and 60 grow maize. If each farmer grows at least one of the two crops, the number who grow both crops is
- A. 10
 - B. 20
 - C. 50
 - D. 30
- 29.** In a group of 60 people, 35 speak English, 30 speak French and 12 speak both languages. The number who speak neither language is
- A. 13
 - B. 17
 - C. 12
 - D. 7
- 31.** In a survey of 100 students, 55 study Mathematics, 45 study Science and 40 study English; 25 study Mathematics and Science, 20 study Science and English, 15 study Mathematics and English, and 10 study all three subjects. The number who study Mathematics only is
- A. 55
 - B. 15
 - C. 25
 - D. 35
- 33.** From the survey in question 31, the number of students who study none of the three subjects is
- A. 90
 - B. 10
 - C. 20
 - D. 0
- 35.** In a survey of 120 people about drinks, 60 take tea, 50 take coffee and 40 take Milo; 20 take tea and coffee, 15 take coffee and Milo, 10 take tea and Milo, and 5 take all three. The number who take tea and coffee but not Milo is
- A. 10
 - B. 25
 - C. 20
 - D. 15
- 28.** From the information in question 27, the number of farmers who grow only one crop is
- A. 50
 - B. 30
 - C. 20
 - D. 80
- 30.** From the information in question 29, the number of people who speak exactly one of the two languages is
- A. 18
 - B. 41
 - C. 53
 - D. 35
- 32.** From the survey in question 31, the number of students who study at least one of the three subjects is
- A. 90
 - B. 80
 - C. 75
 - D. 100
- 34.** From the survey in question 31, the number of students who study exactly two subjects is
- A. 60
 - B. 50
 - C. 30
 - D. 20
- 36.** From the survey in question 35, the number of people who take only Milo is
- A. 20
 - B. 25
 - C. 40
 - D. 15

37. From the survey in question 35, the number of people who take at least one of the drinks is

- A. 120
- B. 100
- C. 110
- D. 105

39. A Venn diagram for three sets divides the universal set into how many regions in all?

- A. 8
- B. 3
- C. 6
- D. 7

38. From the survey in question 35, a person is chosen at random. The probability that the person takes none of the three drinks is

- A. $\frac{11}{12}$
- B. $\frac{1}{12}$
- C. $\frac{1}{6}$
- D. $\frac{1}{10}$

- 40.** For any three sets A, B and C, $n(A \cup B \cup C)$ is equal to
- A. $n(A) + n(B) + n(C) - n(A \cap B \cap C)$
 - B. $n(A) + n(B) + n(C)$
 - C. $n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C)$
 - D. $n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) The number system.

Question 2

- (a) Sets of factors.

Question 3

- (a) A two-set survey.

Question 4

- (a) A three-set survey.

Question 5

- (a) Three sets with an unknown.

Question 6

- (a) Design your own survey.

Chapter 12: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. 4.73×100 is equal to
 - A. 47.3
 - B. 473
 - C. 4,730
 - D. 0.0473
2. $85 \div 1,000$ is equal to
 - A. 8.5
 - B. 0.0085
 - C. 0.085
 - D. 0.85
3. 0.09×10^3 is equal to
 - A. 90
 - B. 0.9
 - C. 9
 - D. 900
4. $620 \div 10^2$ is equal to
 - A. 62
 - B. 0.62
 - C. 6,200
 - D. 6.2
5. The benchmark fraction equivalent to 0.25 is
 - A. $\frac{2}{5}$
 - B. $\frac{1}{4}$
 - C. $\frac{1}{2}$
 - D. $\frac{1}{5}$
6. $\frac{3}{5}$ written as a decimal is
 - A. 0.6
 - B. 0.53
 - C. 0.35
 - D. 3.5
7. $12.5 \times 1,000$ is equal to
 - A. 1,250
 - B. 125,000
 - C. 125
 - D. 12,500
8. $7 \div 100$ is equal to
 - A. 70
 - B. 0.007
 - C. 0.07
 - D. 0.7
9. 0.375 written as a benchmark fraction is
 - A. $\frac{1}{3}$
 - B. $\frac{3}{4}$
 - C. $\frac{5}{8}$
 - D. $\frac{3}{8}$
10. Multiplying a number by 10^{-2} has the same effect as
 - A. dividing the number by 100
 - B. multiplying the number by 100
 - C. dividing the number by 2
 - D. multiplying the number by 2
11. Which of the following illustrates the commutative property of addition?
 - A. $8 + 0 = 8$
 - B. $8 \times 1 = 8$
 - C. $8 + 3 = 3 + 8$
 - D. $(8 + 3) + 2 = 8 + (3 + 2)$
12. $17 \times 8 = 8 \times 17$ is equal to
 - A. 153
 - B. 136
 - C. 126
 - D. 146
13. Which of the following operations is NOT commutative?
 - A. Addition
 - B. Multiplication

- C. Both addition and multiplication
- D. Subtraction

- 14.** If $23 + 48 = 48 + 23$, the property illustrated is the commutative property of
- A. subtraction
 - B. division
 - C. addition

- D. multiplication
- 15.** A student wrote $7 \times 9 = 9 \times 7$. The property used is the
- A. commutative property of addition
 - B. commutative property of multiplication
 - C. associative property of multiplication
 - D. distributive property
- 17.** $(8 + 14) + 22 = 8 + (14 + 22)$ illustrates the associative property of
- A. commutative property of addition
 - B. addition
 - C. distributive property
 - D. multiplication
- 19.** Which of the following illustrates the associative property of multiplication?
- A. $2 \times 3 = 3 \times 2$
 - B. $2 \times (3 + 5) = (2 \times 3) + (2 \times 5)$
 - C. $(2 \times 3) \times 5 = 2 \times (3 \times 5)$
 - D. $2 \times 5 = 5 \times 2$
- 21.** By the associative property of multiplication, $(a \times b) \times c =$
- A. $a \times (b \times c)$
 - B. $(a + b) \times c$
 - C. $(a \times b) + c$
 - D. $a + (b \times c)$
- 23.** $5 \times (10 + 7) = (5 \times 10) + (5 \times 7)$ is equal to
- A. 120
 - B. 75
 - C. 57
 - D. 85
- 25.** Using the distributive property, $6 \times 28 = 6 \times (30 - 2)$ is equal to
- A. 158
 - B. 178
 - C. 162
 - D. 168
- 27.** Using the distributive property, $13 \times 99 = 13 \times (100 - 1)$ is equal to
- A. 1,313
 - B. 1,287
 - C. 1,297
 - D. 1,307
- 16.** By the commutative property of multiplication, $a \times b =$
- A. $b \times a$
 - B. $a + b$
 - C. $a \div b$
 - D. $b \div a$
- 18.** $(5 \times 6) \times 2 = 5 \times (6 \times 2)$ is equal to
- A. 60
 - B. 30
 - C. 50
 - D. 62
- 20.** $(15 + 6) + 9 = 15 + (6 + 9)$ is equal to
- A. 39
 - B. 24
 - C. 21
 - D. 30
- 22.** Using the associative property, the easiest way to evaluate $(25 \times 17) \times 4$ gives
- A. 1,070
 - B. 170
 - C. 1,700
 - D. 1,600
- 24.** $5 \times (10 - 7) = (5 \times 10) - (5 \times 7)$ is equal to
- A. 50
 - B. 15
 - C. 35
 - D. 53
- 26.** $7 \times 46 = (7 \times 40) + (7 \times 6)$ is equal to
- A. 286
 - B. 332
 - C. 322
 - D. 316

D. 1,277

29. Which of the following states the distributive property?

- A. $a + b = b + a$
- B. $a \times (b + c) = (a \times b) + (a \times c)$
- C. $a \times (b + c) = (a + b) \times (a + c)$
- D. $(a \times b) \times c = a \times (b \times c)$

28. $6 \times (20 + 3) = (6 \times 20) + (6 \times 3)$ is equal to

- A. 138
- B. 123
- C. 128
- D. 148

- 31.** Mentally, 0.5×36 is equal to
A. 17
B. 19
C. 16
D. 18
- 33.** Using a mental strategy, 48×25 is equal to
A. 1,100
B. 1,200
C. 1,080
D. 1,400
- 35.** Using compensation, $132 - 98$ is equal to
A. 34
B. 24
C. 36
D. 44
- 37.** By grouping to make tens, $19 + 27 + 31 + 23$ is equal to
A. 100
B. 90
C. 110
D. 99
- 39.** Using a mental strategy, 250×16 is equal to
A. 3,600
B. 2,500
C. 4,000
D. 4,500
- 30.** Using the distributive property, $9 \times (50 - 4)$ is equal to
A. 414
B. 450
C. 404
D. 424
- 32.** $15 \times 12 + 15 \times 8 = 15 \times (12 + 8)$ is equal to
A. 280
B. 315
C. 300
D. 200
- 34.** Using compensation, $49 + 38$ is equal to
A. 77
B. 97
C. 87
D. 88
- 36.** Mentally, 76×5 is equal to
A. 350
B. 385
C. 760
D. 380
- 38.** If $16 \times 7 = 112$, then 16×14 is equal to
A. 212
B. 252
C. 126
D. 224
- 40.** Amina buys 8 exercise books at GH¢4.50 each and pays with a GH¢50 note. Her change is
A. GH¢4.00
B. GH¢14.00
C. GH¢24.00
D. GH¢8.50

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Powers of 10 and benchmark fractions.

Question 2

- (a) The commutative and associative properties.

Question 3

- (a) The distributive property.

Question 4

- (a) Mental mathematics strategies.

Question 5

- (a) Applications.

Question 6

- (a) Reasoning with properties.

Chapter 13: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Evaluate: $4.6 + 3.75$
A. 7.81
B. 8.21
C. 4.975
D. 8.35
2. Evaluate: $9.2 - 3.86$
A. 5.34
B. 5.44
C. 6.34
D. 6.66
3. Evaluate: $0.58 + 0.047$
A. 0.99
B. 0.105
C. 0.627
D. 0.605
4. Evaluate: $7 - 2.46$
A. 5.54
B. 4.54
C. 4.46
D. 5.46
5. Evaluate: $12.35 + 8.9 + 3.06$
A. 24.41
B. 25.31
C. 23.31
D. 24.31
6. Evaluate: $15.06 - 7.48$
A. 8.58
B. 8.42
C. 7.58
D. 7.42
7. Evaluate: $0.9 - 0.009$
A. 0.899
B. 0.891
C. 0.909
D. 0.9
8. Evaluate: $3.75 + 2.5 + 0.25$
A. 6.5
B. 6.55
C. 8.75
D. 6.05
9. Evaluate: $10.005 - 4.7$
A. 5.295
B. 5.705
C. 5.305
D. 6.295
10. Ama spent GH¢18.45 on food and GH¢6.75 on transport. The total amount she spent is
A. GH¢24.20
B. GH¢25.20
C. GH¢25.02
D. GH¢24.85
11. Evaluate: 3.6×7
A. 252
B. 24.2
C. 2.52
D. 25.2
12. Evaluate: 0.25×0.4
A. 0.1
B. 0.65
C. 1.0
D. 0.01
13. Evaluate: $8.4 \div 4$
A. 21
B. 2.1
C. 0.21
D. 2.4
14. Evaluate: $6.3 \div 0.9$
A. 70
B. 5.67
C. 0.7
D. 7

- 15.** Evaluate: 2.5×2.5
A. 5.25
B. 5.0
C. 6.25
D. 62.5
- 17.** Evaluate: 0.12×0.3
A. 3.6
B. 0.036
C. 0.0036
D. 0.36
- 19.** Evaluate: 7.2×0.05
A. 3.6
B. 0.036
C. 36
D. 0.36
- 21.** The value of 3.6×4.7 , correct to 1 decimal place, is
A. 16.8
B. 17.0
C. 16.92
D. 16.9
- 23.** The value of 5.678×1.4 , correct to 2 decimal places, is
A. 7.95
B. 7.949
C. 8.05
D. 7.94
- 25.** The value of $8.91 - 2.348$, correct to 2 decimal places, is
A. 6.57
B. 6.562
C. 6.56
D. 6.66
- 27.** The value of $23.2 \div 3.4$, correct to 2 decimal places, is
A. 6.72
B. 6.83
C. 6.824
D. 6.82
- 29.** A trader sells oranges from two baskets. Basket A contained 85 oranges and she sold 48 of them. The number of oranges left in basket A is
A. 133
B. 38
C. 37
D. 47
- 16.** Evaluate: $9.36 \div 12$
A. 0.78
B. 7.8
C. 0.87
D. 0.078
- 18.** Evaluate: $4.8 \div 0.6$
A. 2.88
B. 0.8
C. 8
D. 80
- 20.** Evaluate: $14.4 \div 1.2$
A. 12
B. 120
C. 13.2
D. 1.2
- 22.** The value of $19.87 \div 6$, correct to 2 decimal places, is
A. 3.32
B. 3.30
C. 3.31
D. 3.312
- 24.** The value of $25.6 \div 7$, correct to 3 significant figures, is
A. 3.7
B. 3.66
C. 3.65
D. 3.657
- 26.** The value of $(4.6 \times 7) \div 9$, correct to 2 decimal places, is
A. 3.577
B. 3.58
C. 3.6
D. 3.57
- 28.** A rope 12.5 m long is cut into 6 equal pieces. The length of each piece, correct to 2 decimal places, is
A. 2.08 m
B. 2.09 m
C. 2.083 m
D. 2.18 m
- 30.** From the information in question 29: the trader sold 59 oranges from basket B and was left with the same number as remained in basket A. The number of oranges originally in basket B was
A. 37

- B. 96
- C. 86
- D. 106

- 31.** Kofi bought rice for GH¢145.75 and cooking oil for GH¢68.40. The total cost of the items is
- A. GH¢213.15
 - B. GH¢204.15
 - C. GH¢214.25
 - D. GH¢214.15
- 33.** One bottle holds 0.75 litre of juice. The total volume of juice in 8 such bottles is
- A. 60 litres
 - B. 6 litres
 - C. 5.6 litres
 - D. 6.75 litres
- 35.** Adjoa runs 2.35 km every morning. The total distance she runs in 5 days is
- A. 11.75 km
 - B. 12.25 km
 - C. 11.35 km
 - D. 10.75 km
- 37.** A rectangle has length 8.4 cm and width 3.25 cm. Its perimeter is
- A. 23.3 cm
 - B. 27.3 cm
 - C. 11.65 cm
 - D. 22.9 cm
- 39.** A minibus used 12.8 litres of fuel over 4 days, the same amount each day. The fuel used per day is
- A. 4.8 litres
 - B. 3.02 litres
 - C. 32 litres
 - D. 3.2 litres
- 32.** From the information in question 31, if Kofi paid with GH¢300, his change is
- A. GH¢85.85
 - B. GH¢85.75
 - C. GH¢86.15
 - D. GH¢95.85
- 34.** 9.6 kg of sugar is shared equally into 4 bags. The mass of sugar in each bag is
- A. 2.04 kg
 - B. 24 kg
 - C. 2.4 kg
 - D. 3.6 kg
- 36.** From the information in question 35, the total distance Adjoa runs in one week (7 days) is
- A. 17.45 km
 - B. 16.10 km
 - C. 15.45 km
 - D. 16.45 km
- 38.** The area of the rectangle in question 37 is
- A. 27.03 cm²
 - B. 11.65 cm²
 - C. 27.3 cm²
 - D. 23.3 cm²
- 40.** Ama has a mass of 42.6 kg and her younger brother has a mass of 38.85 kg. The difference in their masses is
- A. 4.25 kg
 - B. 3.75 kg
 - C. 3.25 kg
 - D. 4.75 kg

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Adding and subtracting decimals.

Question 2

(a) Multiplying and dividing decimals.

Question 3

(a) Mixed operations (use the order of operations).

Question 4

(a) The egg-seller's problem.

Question 5

(a) Shopping with rounding.

Question 6

(a) Create and solve.

Chapter 14: Ch5

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following is a surd?
A. $\sqrt{4}$
B. $\sqrt{16}$
C. $\sqrt{2}$
D. $\sqrt{36}$
2. Evaluate $\sqrt{16}$.
A. 2
B. 36
C. $\sqrt{8}$
D. 4
3. Which of the following is a compound surd?
A. $\sqrt{2}$
B. $2\sqrt{5}$
C. $\sqrt{3} + \sqrt{7} - \sqrt{5}$
D. $7\sqrt{3}$
4. Which of the following surds are like surds?
A. $\sqrt{2}$ and $\sqrt{3}$
B. $2\sqrt{5}$ and $\sqrt{45}$
C. $\sqrt{3}$ and $\sqrt{27}$
D. $\sqrt{8}$ and $\sqrt{18}$
5. The square root of a number which is not a perfect square is
A. a surd
B. always a fraction
C. an integer
D. a whole number
6. Which of the following is NOT a surd?
A. $\sqrt{11}$
B. $\sqrt{49}$
C. $\sqrt{5}$
D. $\sqrt{7}$
7. Simplify $\sqrt{8}$.
A. $2\sqrt{2}$
B. $4\sqrt{2}$
C. 8
D. $2\sqrt{3}$
8. Simplify $\sqrt{12}$.
A. 6
B. $4\sqrt{3}$
C. $3\sqrt{2}$
D. $2\sqrt{3}$
9. Simplify $\sqrt{20}$.
A. $5\sqrt{2}$
B. $2\sqrt{5}$
C. 10
D. $4\sqrt{5}$
10. Simplify $\sqrt{27}$.
A. $9\sqrt{3}$
B. $3\sqrt{9}$
C. 27
D. $3\sqrt{3}$
11. Simplify $\sqrt{45}$.
A. $3\sqrt{5}$
B. $9\sqrt{5}$
C. 5
D. $5\sqrt{3}$
12. Simplify $\sqrt{48}$.
A. $2\sqrt{6}$
B. 24
C. $6\sqrt{2}$
D. $4\sqrt{3}$
13. Simplify $\sqrt{50}$.
A. $10\sqrt{5}$
B. $25\sqrt{2}$
C. $5\sqrt{2}$
D. $2\sqrt{5}$
14. Simplify $\sqrt{72}$.
A. $2\sqrt{6}$
B. $8\sqrt{3}$
C. 36
D. $6\sqrt{2}$

15. Simplify $\sqrt{75}$.

- A. 15
- B. $3\sqrt{5}$
- C. $5\sqrt{3}$
- D. $25\sqrt{3}$

17. Simplify $3\sqrt{2} + 5\sqrt{2}$.

- A. $8\sqrt{10}$
- B. 35
- C. $8\sqrt{2}$
- D. $15\sqrt{2}$

19. Simplify $7\sqrt{2} + 3\sqrt{2}$.

- A. $10\sqrt{10}$
- B. $10\sqrt{2}$
- C. 21
- D. 10

21. Simplify $4\sqrt{7} + 5\sqrt{7}$.

- A. $\sqrt{7}$
- B. $9\sqrt{7}$
- C. $9\sqrt{35}$
- D. $20\sqrt{7}$

23. Simplify $7\sqrt{2} - 2\sqrt{2}$.

- A. $\sqrt{2}$
- B. $5\sqrt{2}$
- C. 14
- D. $9\sqrt{2}$

25. Simplify $\sqrt{18} + \sqrt{8}$.

- A. $5\sqrt{2}$
- B. $\sqrt{5}$
- C. 10
- D. $5\sqrt{10}$

27. Simplify $3\sqrt{3} + 4\sqrt{3}$.

- A. $\sqrt{3}$
- B. $7\sqrt{3}$
- C. $12\sqrt{3}$
- D. 7

29. Simplify $\sqrt{12} + \sqrt{27}$.

- A. $7\sqrt{5}$
- B. $\sqrt{15}$
- C. $5\sqrt{3}$
- D. $5\sqrt{15}$

31. Evaluate $\sqrt{2} \times \sqrt{3}$.

- A. $\sqrt{6}$
- B. $\sqrt{5}$
- C. 6
- D. $\sqrt{23}$

16. Simplify $\sqrt{200}$.

- A. $10\sqrt{2}$
- B. 100
- C. 20
- D. $2\sqrt{10}$

18. Simplify $9\sqrt{5} - 4\sqrt{5}$.

- A. $5\sqrt{5}$
- B. 5
- C. $5\sqrt{10}$
- D. $\sqrt{5}$

20. Simplify $6\sqrt{3} + 4\sqrt{3}$.

- A. $\sqrt{3}$
- B. $10\sqrt{9}$
- C. $10\sqrt{3}$
- D. 10

22. Simplify $3\sqrt{5} + 2\sqrt{5} - \sqrt{5}$.

- A. $6\sqrt{5}$
- B. $\sqrt{5}$
- C. $4\sqrt{5}$
- D. 0

24. Which of the following is equal to $5\sqrt{2}$?

- A. $\sqrt{45}$
- B. $\sqrt{20}$
- C. $\sqrt{12}$
- D. $\sqrt{50}$

26. Simplify $3\sqrt{2} + 2\sqrt{2}$.

- A. $\sqrt{2}$
- B. 12
- C. $5\sqrt{2}$
- D. $6\sqrt{2}$

28. Simplify $2\sqrt{2} + 5\sqrt{2}$.

- A. 10
- B. $\sqrt{10}$
- C. $7\sqrt{20}$
- D. $7\sqrt{2}$

30. Simplify $\sqrt{3} + \sqrt{5}$.

- A. $2\sqrt{2}$
- B. $\sqrt{8}$
- C. $\sqrt{15}$
- D. $\sqrt{3} + \sqrt{5}$

32. Evaluate $\sqrt{5} \times \sqrt{5}$.

- A. 25
- B. 10
- C. $\sqrt{10}$
- D. 5

33. Evaluate $2\sqrt{3} \times \sqrt{2}$.

- A. $2\sqrt{6}$
- B. 12
- C. $6\sqrt{6}$
- D. $\sqrt{6}$

35. Evaluate $\sqrt{8} \times \sqrt{2}$.

- A. 16
- B. 8
- C. 2
- D. 4

37. Simplify $\sqrt{12} \div \sqrt{3}$.

- A. 4
- B. $\sqrt{9}$
- C. 2
- D. 12

39. Simplify $\sqrt{18} \div \sqrt{2}$.

- A. $\sqrt{9}$
- B. 3
- C. 9
- D. $\sqrt{16}$

34. Evaluate $3\sqrt{5} \times 2\sqrt{5}$.

- A. $6\sqrt{5}$
- B. 30
- C. 15
- D. $30\sqrt{5}$

36. Evaluate $\sqrt{7} \times \sqrt{28}$.

- A. 98
- B. 14
- C. $\sqrt{35}$
- D. 7

38. Simplify $\sqrt{50} \div \sqrt{2}$.

- A. 5
- B. $\sqrt{48}$
- C. 10
- D. 25

40. Simplify $(6\sqrt{2}) \div (3\sqrt{2})$.

- A. 2
- B. 3
- C. $2\sqrt{2}$
- D. 6

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Simplifying surds

Question 2

(a) Adding and subtracting surds

Question 3

(a) Multiplying and dividing surds

Question 4

(a) Surds in context

Question 5

(a) Identities and exact values

Question 6

(a) Create and solve

Chapter 15: Ch6

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following fractions is equivalent to $\frac{2}{5}$?
A. $\frac{3}{10}$
B. $\frac{5}{2}$
C. $\frac{2}{10}$
D. $\frac{4}{10}$
2. Express $\frac{15}{10}$ in its simplest form.
A. $\frac{3}{2}$
B. $\frac{1}{2}$
C. $\frac{5}{10}$
D. $\frac{5}{3}$
3. Express $\frac{12}{5}$ as a mixed number.
A. $2\frac{3}{5}$
B. $2\frac{1}{5}$
C. $1\frac{2}{5}$
D. $2\frac{2}{5}$
4. Express $2\frac{5}{9}$ as an improper fraction.
A. $\frac{23}{9}$
B. $\frac{19}{9}$
C. $\frac{14}{9}$
D. $\frac{25}{9}$
5. Which of the following fractions is the greatest?
A. $\frac{5}{8}$
B. $\frac{3}{4}$
C. $\frac{7}{12}$
D. $\frac{2}{3}$
6. Which of the following fractions lies between $\frac{1}{4}$ and $\frac{1}{2}$?
A. $\frac{3}{4}$
B. $\frac{1}{8}$
C. $\frac{5}{8}$
D. $\frac{3}{8}$
7. Reduce $\frac{18}{24}$ to its lowest terms.
A. $\frac{2}{3}$
B. $\frac{9}{16}$
C. $\frac{3}{4}$
D. $\frac{4}{3}$
8. Express 0.75 as a fraction in its lowest terms.
A. $\frac{3}{5}$
B. $\frac{3}{4}$
C. $\frac{7}{25}$
D. $\frac{7}{10}$
9. Work out $\frac{3}{4} + \frac{7}{8}$.
A. $1\frac{5}{8}$
B. $1\frac{3}{8}$
C. $\frac{10}{12}$
D. $1\frac{1}{8}$
10. Work out $\frac{1}{2} + \frac{1}{3}$.
A. $\frac{2}{5}$
B. $\frac{1}{6}$
C. $\frac{2}{6}$
D. $\frac{5}{6}$
11. Work out $\frac{5}{6} - \frac{1}{4}$.
A. $\frac{2}{3}$
B. $\frac{1}{12}$
C. $\frac{7}{12}$
D. $\frac{1}{3}$
12. Work out $1\frac{1}{2} + \frac{4}{5} - \frac{5}{6}$.
A. $1\frac{4}{5}$
B. $1\frac{7}{15}$
C. $1\frac{2}{15}$
D. $\frac{11}{15}$
13. Work out $2\frac{1}{4} - 1\frac{1}{2}$.
A. $1\frac{1}{4}$
B. $\frac{1}{2}$
C. $\frac{3}{4}$
D. $1\frac{3}{4}$
14. Work out $\frac{3}{8} + \frac{1}{6}$.
A. $\frac{13}{24}$
B. $\frac{5}{24}$
C. $\frac{11}{24}$
D. $\frac{4}{14}$

15. Work out $7/9 - 1/6$.

- A. $5/18$
- B. $1/3$
- C. $11/18$
- D. $1/2$

17. Evaluate $2/3 \times 3/4$.

- A. $8/9$
- B. $1/4$
- C. $1/2$
- D. $5/7$

19. Find $3/5$ of 40.

- A. 15
- B. 24
- C. 32
- D. 8

21. Evaluate $2 \frac{1}{2} \div 1/4$.

- A. 4
- B. $5/8$
- C. 10
- D. $9/8$

23. Evaluate $7/8 \div 7$.

- A. $1/56$
- B. 8
- C. $49/8$
- D. $1/8$

25. Using BODMAS, evaluate $2/3 \times 3/4 - 5/8 \div 2 \frac{1}{2}$.

- A. $1/8$
- B. $1/4$
- C. $3/8$
- D. $1/2$

27. Using BODMAS, evaluate $(3/4 + 5/8) \times 4/11 - 1/2$.

- A. 2
- B. 1
- C. $1/2$
- D. 0

29. Evaluate $(2/3 + 1/6) \div 5/12$.

- A. 2
- B. $5/2$
- C. $1/2$
- D. 1

31. Evaluate $4 - 1 \frac{1}{2} \div 3/4$.

- A. $10/3$
- B. 1
- C. 3

16. What must be added to $3/7$ to give 1?

- A. $3/7$
- B. $4/7$
- C. $7/3$
- D. $1/7$

18. Evaluate $5/6 \div 2/3$.

- A. $3/10$
- B. $5/9$
- C. $4/5$
- D. $1 \frac{1}{4}$

20. Evaluate $4/9 \times 3/8$.

- A. $1/6$
- B. $1/2$
- C. $2/3$
- D. $7/17$

22. Evaluate $(3/4)^2$.

- A. $3/2$
- B. $9/16$
- C. $3/8$
- D. $9/8$

24. Evaluate $5/8 \div 2 \frac{1}{2} + 2/3$.

- A. $11/12$
- B. $1/6$
- C. $1/4$
- D. $13/12$

26. Using BODMAS, evaluate $3/4 \div 3/8 + (4/5 - 1/2)$.

- A. $8/5$
- B. $1 \frac{3}{10}$
- C. $2 \frac{3}{10}$
- D. $2 \frac{1}{10}$

28. Evaluate $1/2 + 1/4 \times 2/3$.

- A. $1/2$
- B. $1/4$
- C. $2/3$
- D. $5/6$

30. Evaluate $3/5 \times (1/2 + 1/3)$.

- A. $5/6$
- B. $1/2$
- C. $3/10$
- D. $1/5$

D. 2

- 33.** A test is made up of 20 questions.
How many questions must you
answer correctly to get a score of
80%?

A. 16
B. 18
C. 12
D. 8

- 32.** Find $\frac{2}{3}$ of $\frac{1}{4}$ of 60.

A. 20
B. 5
C. 10
D. 40

- 35.** In an election, 5,000 people voted. One candidate claimed 52% of the votes while the other claimed 2,681 votes. What is the maximum number of votes the second candidate could actually have received?
- A. 2,681
 - B. 2,500
 - C. 2,400
 - D. 2,600
- 37.** Ama has GH¢120. She spends $\frac{1}{3}$ of it on food and $\frac{1}{4}$ on transport. How much is left?
- A. GH¢70
 - B. GH¢40
 - C. GH¢110
 - D. GH¢50
- 39.** $\frac{5}{7}$ of a class of 42 pupils are girls. How many boys are in the class?
- A. 21
 - B. 30
 - C. 7
 - D. 12
- 34.** A television set was marked GH¢2,250 and sold for GH¢2,025. By what percentage was it reduced?
- A. 90%
 - B. 10%
 - C. 225%
 - D. 25%
- 36.** Esi's drink is made of $\frac{3}{8}$ orange squash and Fusena's of $\frac{2}{5}$ orange squash. Whose drink tastes stronger of orange?
- A. Fusena's drink
 - B. Cannot be determined
 - C. Esi's drink
 - D. Both taste equally strong
- 38.** A rope $7\frac{1}{2}$ m long is cut into pieces each $1\frac{1}{2}$ m long. How many pieces are obtained?
- A. 5
 - B. 6
 - C. 3
 - D. 4
- 40.** Express $\frac{2}{3}$ as a decimal, correct to 2 decimal places.
- A. 0.66
 - B. 0.67
 - C. 0.6
 - D. 0.7

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Review of fractions

Question 2

- (a) Addition and subtraction of fractions

Question 3

- (a) Multiplication, division and BODMAS

Question 4

- (a) Fractions of quantities

Question 5

- (a) Word problems with rounding

Question 6

- (a) Create and solve

Chapter 16: Ch

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Express 12 : 18 in its simplest form.
A. 3 : 2
B. 2 : 3
C. 4 : 6
D. 12 : 18
2. GH¢120 is shared between Ama and Kofi in the ratio 2 : 3. How much does Ama receive?
A. GH¢48
B. GH¢72
C. GH¢60
D. GH¢50
3. In a class, the ratio of boys to girls is 3 : 5. If there are 40 pupils, how many boys are there?
A. 12
B. 15
C. 25
D. 24
4. A car travels 180 km in 3 hours. Find its speed.
A. 540 km/h
B. 90 km/h
C. 45 km/h
D. 60 km/h
5. If 4 pens cost GH¢12, find the cost of 7 pens.
A. GH¢14
B. GH¢21
C. GH¢24
D. GH¢28
6. The total cost t of items is proportional to the number n at GH¢4.50 each ($t = pn$). Find t when $n = 6$.
A. GH¢27.00
B. GH¢13.50
C. GH¢10.50
D. GH¢27.50
7. The graph of two quantities is a straight line through the origin. The relationship is
A. inversely proportional
B. constant
C. directly proportional
D. zero
8. How much do 8 avocados cost?
A. GH¢10.50
B. GH¢12.00
C. GH¢16.00
D. GH¢8.00
9. Using the graph, how much do 15 avocados cost?
A. GH¢24.00
B. GH¢15.00
C. GH¢20.00
D. GH¢22.50
10. The unit rate of the graph (its slope) is
A. GH¢1.50 per avocado
B. GH¢3.00 per avocado
C. GH¢2.00 per avocado
D. GH¢0.50 per avocado
11. In the simple interest formula $I = \frac{PRT}{100}$, the letter R represents the
A. time in years
B. amount
C. principal
D. rate of interest per annum
12. Find the simple interest on GH¢4,500 invested at 3% per annum for 3 years.
A. GH¢4,905
B. GH¢405
C. GH¢135

D. GH¢450

- 13.** The amount at the end of the 3 years in the previous question is
- A. GH¢4,905
 - B. GH¢4,505
 - C. GH¢4,635
 - D. GH¢5,175
- 15.** Find the amount on GH¢2,000 invested at 5% per annum simple interest for 2 years.
- A. GH¢2,500
 - B. GH¢2,100
 - C. GH¢2,200
 - D. GH¢2,020
- 17.** The VAT rate in Ghana is 12.5%. The VAT on an item priced GH¢400 (exclusive) is
- A. GH¢450
 - B. GH¢500
 - C. GH¢12.50
 - D. GH¢50
- 19.** The VAT paid by the man is
- A. GH¢562.50
 - B. GH¢450
 - C. GH¢625
 - D. GH¢500
- 21.** The discount allowed on the car is
- A. GH¢10,050
 - B. GH¢8,500
 - C. GH¢8,050
 - D. GH¢805
- 23.** A mobile phone originally worth GH¢1,800 depreciates by 5% of its original value in the first year. Its value at the end of the first year is
- A. GH¢1,755
 - B. GH¢1,710
 - C. GH¢90
 - D. GH¢1,620
- 25.** The NHIL inclusive price of a television set is GH¢1,200.00. If the NHIL is charged at 2.5%, the cost of the television (NHIL exclusive) is
- A. GH¢1,170.73
 - B. GH¢1,170.00
 - C. GH¢30.00
 - D. GH¢1,230.00
- 14.** The amount at the end of the fifth year is
- A. GH¢5,025
 - B. GH¢5,850
 - C. GH¢4,905
 - D. GH¢5,175
- 16.** The simple interest on a principal for 4 years at 5% per annum is GH¢300. The principal is
- A. GH¢60
 - B. GH¢1,500
 - C. GH¢3,000
 - D. GH¢1,200
- 18.** A man bought an item at GH¢4,500.00, VAT inclusive, at 12.5%. The basic cost of the item is
- A. GH¢4,500
 - B. GH¢4,000
 - C. GH¢3,937.50
 - D. GH¢4,062.50
- 20.** A car costs GH¢80,500.00. Its new value after a discount of 10% is
- A. GH¢72,450
 - B. GH¢72,500
 - C. GH¢70,450
 - D. GH¢79,695
- 22.** An agent's commission on the sale of a car is $3\frac{1}{2}\%$. The commission on a car sold for GH¢68,000.00 is
- A. GH¢238
 - B. GH¢2,040
 - C. GH¢3,400
 - D. GH¢2,380
- 24.** In the second year the phone depreciates by 10% of the original value. Its value at the end of the second year is
- A. GH¢1,710
 - B. GH¢1,530
 - C. GH¢1,620
 - D. GH¢1,638
- 26.** The NHIL charged is
- A. GH¢25.00
 - B. GH¢29.00
 - C. GH¢29.27
 - D. GH¢30.00

- 27.** Kofi Mereku paid an insurance premium of GH¢30,000.00. If the premium is 5% of the value of the house, the insured value is
- A. GH¢60,000
 - B. GH¢150,000
 - C. GH¢1,500,000
 - D. GH¢600,000
- 29.** Under Act 766, the SSNIT contribution rate of the employer is
- A. 12.5% of basic salary
 - B. 17.5% of basic salary
 - C. 5.5% of basic salary
 - D. 13% of basic salary
- 31.** A worker's basic monthly salary is GH¢3,256.50. The worker's SSNIT contribution under Act 766 is
- A. GH¢423.35
 - B. GH¢162.83
 - C. GH¢179.11
 - D. GH¢179.00
- 33.** The total SSNIT contribution at the end of every month is
- A. GH¢625.00
 - B. GH¢602.46
 - C. GH¢602.45
 - D. GH¢586.17
- 35.** Mr Bediako's monthly SSNIT contribution under PNDCL 247 is GH¢440.54. His employer contributes
- A. GH¢440.54
 - B. GH¢1,221.50
 - C. GH¢1,101.35
 - D. GH¢881.08
- 37.** If $3a = 4b$, then $a : b$ is
- A. 3 : 4
 - B. 12 : 1
 - C. 1 : 12
 - D. 4 : 3
- 39.** 5 workers take 12 days to complete a job. At the same rate, 10 workers will take
- A. 4 days
 - B. 24 days
 - C. 6 days
 - D. 9 days
- 28.** Under Act 766, the SSNIT contribution rate of the worker is
- A. 13% of basic salary
 - B. 5% of basic salary
 - C. 5.5% of basic salary
 - D. 18.5% of basic salary
- 30.** The total monthly SSNIT contribution under Act 766 is
- A. 18.5% of basic salary
 - B. 8.5% of basic salary
 - C. 17.5% of basic salary
 - D. 18% of basic salary
- 32.** The employer's SSNIT contribution on the same salary is
- A. GH¢423.34
 - B. GH¢423.35
 - C. GH¢406.56
 - D. GH¢179.11
- 34.** Under PNDCL 247, the worker's SSNIT contribution rate is
- A. 5.5% of basic salary
 - B. 5% of basic salary
 - C. 12.5% of basic salary
 - D. 13% of basic salary
- 36.** Mr Bediako's basic salary per month is
- A. GH¢8,810.80
 - B. GH¢3,524.32
 - C. GH¢4,405.40
 - D. GH¢8,810.00
- 38.** A map has a scale of 1 : 50,000. A road measures 4 cm on the map. The actual length of the road is
- A. 2 km
 - B. 200 m
 - C. 200,000 km
 - D. 20 km
- 40.** y is directly proportional to x , so $y = kx$. If $y = 12$ when $x = 3$, then $k =$
- A. 4
 - B. 36
 - C. 15
 - D. 9

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Ratio and sharing

Question 2

- (a) Rates and proportion

Question 3

- (a) Simple interest

Question 4

- (a) VAT, discount and commission

Question 5

- (a) SSNIT contributions

- (a) 5% of basic salary

- (a) 0% of basic salary

Question 13

- (a) 0% of basic salary

Question 12

- (a) 5% of basic salary

Question 18

- (a) 5% of basic salary

Question 17

- (a) 5% of basic salary

Question 6

- (a) Graphing a proportional relationship

Chapter 17: Ch8

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The value of y when $x = 3$ on the line $y = 2x + 1$ is
 - A. 5
 - B. 6
 - C. 8
 - D. 7
2. The value of y when $x = -3$ on the relation $y = -x + 5$ is
 - A. 8
 - B. -2
 - C. 3
 - D. 2
3. The value of y when $x = 4$ on the relation $y = \frac{1}{2}x - 3$ is
 - A. -2
 - B. 1
 - C. -1
 - D. -5
4. The value of y when $x = -2$ on the line $y = 3x - 2$ is
 - A. -8
 - B. -4
 - C. -6
 - D. 4
5. Which ordered pair lies on the line $y = x + 4$?
 - A. $(-4, 1)$
 - B. $(-1, 3)$
 - C. $(0, -4)$
 - D. $(1, 3)$
6. Which point does NOT lie on the line $y = 2x - 1$?
 - A. $(2, 4)$
 - B. $(0, -1)$
 - C. $(1, 1)$
 - D. $(3, 5)$
7. Completing the table for $y = -x + 5$: when $x = 0$, y is
 - A. 0
 - B. -5
 - C. 5
 - D. 1
8. Completing the table for $y = \frac{1}{2}x - 3$: when $x = -2$, y is
 - A. -1
 - B. -4
 - C. 2
 - D. -2
9. The gradient of the line $y = -x + 4$ is
 - A. -4
 - B. 1
 - C. -1
 - D. 4
10. The y -intercept of the line $y = x - 2$ is
 - A. 2
 - B. 0
 - C. 1
 - D. -2
11. On the line $y = -x + 4$, when $x = -1$, y is
 - A. -5
 - B. -3
 - C. 5
 - D. 3
12. On the line $y = x - 2$, when $x = -1$, y is
 - A. 1
 - B. -3
 - C. -1
 - D. 3

- 13.** The coordinates of the intersection of $y = -x + 4$ and $y = x - 2$ are
A. (1, 3)
B. (2, 2)
C. (3, 1)
D. (-3, 7)
- 15.** The line $y = -x + 4$ meets the y-axis at
A. (0, 4)
B. (1, 4)
C. (4, 0)
D. (0, -4)
- 17.** Solving $y = -x + 7$ and $y = 2x + 1$ simultaneously gives
A. (1, 6)
B. (2, 5)
C. (3, 4)
D. (5, 2)
- 19.** Solving $y = 2x$ and $y = x + 3$ simultaneously gives
A. (6, 3)
B. (3, 6)
C. (3, 5)
D. (2, 4)
- 21.** The lines $y = 2x + 1$ and $y = 2x - 3$ have
A. no solution (they are parallel)
B. the solution (1, 3)
C. exactly one solution
D. infinitely many solutions
- 23.** The graphs of $y = 2x - 1$ and $y = -x + 5$ intersect at
A. (1, 1)
B. (2, 3)
C. (0, 5)
D. (3, 2)
- 25.** Solve simultaneously: $x + y = 7$ and $x - y = 3$.
A. $x = 2, y = 5$
B. $x = 5, y = 2$
C. $x = 4, y = 3$
D. $x = 5, y = 3$
- 27.** Solve simultaneously: $3x - y = 7$ and $x + y = 5$.
A. $x = 1, y = 4$
B. $x = 3, y = 2$
- 14.** The line $y = x - 2$ crosses the x-axis at $x =$
A. 0
B. 2
C. 4
D. -2
- 16.** The gradient of the line $y = \frac{1}{2}x - 3$ is
A. -3
B. 2
C. $\frac{1}{2}$
D. $-\frac{1}{2}$
- 18.** Solving $y = x + 4$ and $y = 6 - x$ simultaneously gives
A. (-1, 5)
B. (5, 1)
C. (2, 4)
D. (1, 5)
- 20.** Solving $y = x + 1$ and $y = 2x - 1$ simultaneously gives
A. (1, 2)
B. (2, 1)
C. (2, 3)
D. (3, 2)
- 22.** The equations $y = x + 2$ and $y = x + 2$ have
A. no solution
B. exactly one solution
C. the solution (0, 2) only
D. infinitely many solutions
- 24.** Using the graph, when $x = 4$ the value of y on $y = 2x - 1$ is
A. 7
B. 3
C. 8
D. 6
- 26.** Solve simultaneously: $2x + y = 8$ and $x + y = 5$.
A. $x = 4, y = 0$
B. $x = 3, y = 3$
C. $x = 2, y = 3$
D. $x = 3, y = 2$

- C. $x = 2, y = 1$
- D. $x = 3, y = -2$

- 29.** A pen costs GH¢2 more than a pencil. Three pens and two pencils together cost GH¢16. The price of a pen is
- A. GH¢4
 - B. GH¢5
 - C. GH¢6
 - D. GH¢3

- 28.** The sum of two numbers is 10 and their difference is 4. The numbers are
- A. 6 and 4
 - B. 8 and 2
 - C. 5 and 5
 - D. 7 and 3

- 31.** Two goats and one sheep cost GH¢700; one goat and two sheep cost GH¢800. The cost of a goat is
A. GH¢300
B. GH¢250
C. GH¢200
D. GH¢350
- 33.** The value of y when $x = 3$ on the relation $y = -x + 5$ is
A. 2
B. 3
C. -2
D. 8
- 35.** The relation shown by the table x : 0, 1, 2, 3; y : 1, 3, 5, 7 is
A. $y = 2x + 1$
B. $y = 2x - 1$
C. $y = 3x + 1$
D. $y = x + 1$
- 37.** The graph of $y = -x + 5$ is
A. a line sloping downwards from left to right, crossing the y -axis at 5
B. a line sloping upwards from left to right, crossing the y -axis at 5
C. a line sloping downwards from left to right, crossing the y -axis at -5
D. a line sloping upwards from left to right, crossing the y -axis at -5
- 39.** The graphs of two relations intersect at (2, 5). The solution of the simultaneous equations is
A. $x = 7$
B. $x = 5$ and $y = 2$
C. $x = 2$ only
D. $x = 2$ and $y = 5$
- 30.** Ama is 3 years older than Kofi. The sum of their ages is 25. Ama's age is
A. 12
B. 13
C. 15
D. 14
- 32.** Solving $y = 3x$ and $y = x + 8$ simultaneously gives
A. (12, 4)
B. (4, 20)
C. (8, 24)
D. (4, 12)
- 34.** The value of y when $x = 0$ on the relation $y = \frac{1}{2}x - 3$ is
A. 3
B. 0
C. -1.5
D. -3
- 36.** Which table of values represents $y = x - 2$?
A. (-2, 0), (0, 2), (2, 4)
B. (0, 2), (2, 4), (4, 6)
C. (0, -2), (2, 0), (4, 2)
D. (0, 2), (2, 0), (4, -2)
- 38.** The line $y = \frac{1}{2}x - 3$ crosses the x -axis at $x =$
A. -6
B. 6
C. -3
D. 3
- 40.** Graphically, the solution of simultaneous linear equations is found at
A. the y -intercepts of the lines
B. the x -intercepts of the lines
C. the intersection of the two lines
D. the midpoint of the lines

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Tables of values and intersection

Question 2

- (a) Graphing two linear relations

Question 3

- (a) Solving simultaneous equations algebraically

Question 4

- (a) Word problems

Question 5

- (a) Reading the graph (Fig. 1)

Question 6

- (a) Create and solve

Chapter 18: Ch

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Make x the subject of the formula $x + a = b$.
 A. $x = a - b$
 B. $x = b - a$
 C. $x = b + a$
 D. $x = -b - a$
2. Make x the subject of the formula $2x = y$.
 A. $x = y - 2$
 B. $x = y + 2$
 C. $x = y/2$
 D. $x = 2y$
3. Make r the subject of the formula $C = 2\pi r$.
 A. $r = C/(2\pi)$
 B. $r = C - 2\pi$
 C. $r = C/\pi$
 D. $r = 2\pi C$
4. Make h the subject of the formula $A = \frac{1}{2}bh$.
 A. $h = A/b$
 B. $h = A/(2b)$
 C. $h = \frac{1}{2}Ab$
 D. $h = 2A/b$
5. Make t the subject of the formula $v = u + at$.
 A. $t = (v - u)/a$
 B. $t = (v - u) \times a$
 C. $t = v - u/a$
 D. $t = (v + u)/a$
6. Make b the subject of the formula $A = l \times b$.
 A. $b = A - l$
 B. $b = A/l$
 C. $b = A \times l$
 D. $b = l/A$
7. Make x the subject of the formula $y = 3x - 2$.
 A. $x = (y - 2)/3$
 B. $x = 3(y + 2)$
 C. $x = (y + 2)/3$
 D. $x = y/3 + 2$
8. Make r the subject of the formula $A = \pi r^2$.
 A. $r = \sqrt{(A\pi)}$
 B. $r = \sqrt{(A/\pi)}$
 C. $r = 2A/\pi$
 D. $r = A/\pi$
9. Make l the subject of the formula $P = 2(l + b)$.
 A. $l = P/2 - b$
 B. $l = 2P - b$
 C. $l = P - 2b$
 D. $l = (P - b)/2$
10. Make a the subject of the formula $v^2 = u^2 + 2as$.
 A. $a = (v - u)/(2s)$
 B. $a = (v^2 - u^2) \times 2s$
 C. $a = (v^2 - u^2)/(2s)$
 D. $a = (v^2 + u^2)/(2s)$
11. The area of a rectangle is 24 cm^2 . If the length is 8 cm , the width is (curriculum exemplar)
 A. 16 cm
 B. 6 cm
 C. 4 cm
 D. 3 cm
12. A circle has an area of 154 cm^2 . Taking $\pi = 22/7$, its radius is (curriculum exemplar)
 A. 14 cm
 B. 7 cm
 C. 49 cm
 D. 22 cm

- 13.** The triangle has an area of 54 cm^2 . The value of the height h is (curriculum exemplar)
- A. 4.5 cm
 - B. 108 cm
 - C. 9 cm
 - D. 12 cm
- 15.** Find the value of $x^2 + 3x$ when $x = 2$.
- A. 8
 - B. 7
 - C. 12
 - D. 10
- 17.** Find the value of $(x - b)^2 - 3(x - b)$ when $x = 2$ and $b = -5$ (curriculum exemplar).
- A. 49
 - B. 21
 - C. 7
 - D. 28
- 19.** Evaluate $(p + q)(p - q)$ when $p = 5$ and $q = 3$.
- A. 8
 - B. 16
 - C. 2
 - D. 34
- 21.** The volume of the shape is $\frac{1}{3}\pi r^2 h$. The volume when $r = 7$, $h = 21$ and $\pi = 22/7$ is (curriculum exemplar)
- A. $1,078 \text{ cm}^3$
 - B. 462 cm^3
 - C. $3,234 \text{ cm}^3$
 - D. 154 cm^3
- 23.** Using $v = u + at$, find v when $u = 5$, $a = 2$ and $t = 4$.
- A. 14
 - B. 13
 - C. 7
 - D. 11
- 25.** Evaluate $(2x + y)/3$ when $x = 5$ and $y = 2$.
- A. 3
 - B. $3 \frac{1}{3}$
 - C. 4
 - D. 6
- 27.** Factorise $3x^2 + 4xy$ (curriculum exemplar).
- A. $xy(3 + 4)$
- 14.** The cylinder has a volume of $1,540 \text{ cm}^3$. Taking $\pi = 22/7$, the value of the height h is
- A. 44 cm
 - B. 10 cm
 - C. 22 cm
 - D. 100 cm
- 16.** Evaluate $2a - 3b$ when $a = 4$ and $b = 1$.
- A. 5
 - B. 11
 - C. -1
 - D. 6
- 18.** Evaluate $3m^2 - 2m + 1$ when $m = -1$.
- A. 6
 - B. 0
 - C. 2
 - D. 4
- 20.** Using $s = \frac{1}{2}gt^2$, find s when $g = 10$ and $t = 3$.
- A. 90
 - B. 30
 - C. 15
 - D. 45
- 22.** Using $I = PRT/100$, find I when $P = 500$, $R = 4$ and $T = 3$.
- A. 600
 - B. 120
 - C. 6,000
 - D. 60
- 24.** Evaluate $a^2 - b^2$ when $a = 7$ and $b = 3$.
- A. 44
 - B. 16
 - C. 4
 - D. 40
- 26.** Evaluate $\sqrt{a^2 + b^2}$ when $a = 3$ and $b = 4$.
- A. 5
 - B. 25
 - C. 1
 - D. 7

- B. $x(4 + 3y)$
- C. $3x(1 + 4y)$
- D. $x(3 + 4y)$

28. Factorise $12ab + 16b$ (curriculum exemplar).

- A. $4a(3b + 4)$
- B. $4b(3a + 4b)$

- C. $4b(3a + 4)$
D. $16b(3a + 1)$
- 29.** Factorise $-13xy + 39x$ (curriculum exemplar).
A. $-13x(y + 3)$
B. $-13x(y - 3)$
C. $-13x(3 - y)$
D. $13x(y - 3)$
- 31.** Factorise $5x + 10$.
A. $5(x + 2)$
B. $5(x + 10)$
C. $x(5 + 2)$
D. $5x + 2$
- 33.** Factorise $6a^2 - 9a$.
A. $6a(a - 9)$
B. $3a(2a - 3)$
C. $3a(2a + 3)$
D. $9a(a - 3)$
- 35.** Factorise $ax + ay + bx + by$.
A. $(a + b)(x + y)$
B. $(a - b)(x + y)$
C. $(a + x)(b + y)$
D. $(a + b)(x - y)$
- 37.** Factorise $7p - 7q$.
A. $7(p - q)$
B. $7(p + q)$
C. $7p - q$
D. $p - 7q$
- 39.** The highest common factor of $6x^2y$ and $9xy^2$ is
A. $6xy$
B. $3xy$
C. xy
D. $3x^2y^2$
- C. $4b(3a + 4)$
D. $16b(3a + 1)$
- 30.** Factorise $8y - 2y^2$ (curriculum exemplar).
A. $y(8 - y)$
B. $2y(4 - 2y)$
C. $2y(4 - y)$
D. $2y(4 + y)$
- 32.** Factorise $x^2 + 3x$.
A. $x^2 + 3$
B. $3x(x + 1)$
C. $x(x + 3)$
D. $(x + 3)$
- 34.** Factorise completely $4x^2 - 16$.
A. $(x + 4)(x - 4)$
B. $4(x - 2)^2$
C. $(2x - 4)(x + 2)$
D. $4(x + 2)(x - 2)$
- 36.** Factorise $2x^2 + 5x + 2$.
A. $(2x + 1)(x - 2)$
B. $(2x - 1)(x - 2)$
C. $(x + 1)(2x + 2)$
D. $(2x + 1)(x + 2)$
- 38.** Factorise $x^2 - 9$.
A. $(x + 9)(x - 1)$
B. $(x + 3)^2$
C. $(x + 3)(x - 3)$
D. $(x - 3)^2$
- 40.** Factorise $15m^2n - 10mn^2$.
A. $5mn(3m + 2n)$
B. $5mn(3 - 2n)$
C. $5mn(3m - 2n)$
D. $10mn(3m - 2n)$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Change of subject

Question 2

(a) Using changed subjects with the figures

Question 3

(a) Substitution

Question 4

(a) Factorisation (curriculum exemplars)

Question 5

(a) Mixed: substitution and factorisation

Question 6

(a) Create and solve

ANSWER KEY

Ch10**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Solve a simple inequality	B9.2.3.1.1
2	C	Divide by a positive number	B9.2.3.1.1
3	A	Solve a simple inequality	B9.2.3.1.1
4	D	Divide by a positive number	B9.2.3.1.1
5	A	Multiply by -1 (sign flips)	B9.2.3.1.1
6	B	Divide by a negative (sign flips)	B9.2.3.1.1
7	C	Rational coefficients (curriculum exemplar)	B9.2.3.1.1
8	A	Two-step inequality	B9.2.3.1.1
9	B	Variable with negative coefficient	B9.2.3.1.1
10	A	Multiply both sides	B9.2.3.1.1
11	C	Rational coefficients	B9.2.3.1.1
12	D	Negative coefficient (sign flips)	B9.2.3.1.1
13	B	Variables on both sides	B9.2.3.1.1
14	A	Expand then solve	B9.2.3.1.1
15	C	Rational coefficients (curriculum exemplar)	B9.2.3.1.1
16	A	Brackets with fractions	B9.2.3.1.1
17	D	Read a number-line graph	B9.2.3.1.2
18	C	Read a number-line graph	B9.2.3.1.2
19	D	Meaning of an open circle	B9.2.3.1.2
20	C	Meaning of a closed circle	B9.2.3.1.2
21	D	Read a number-line graph	B9.2.3.1.2
22	B	Choose the correct graph	B9.2.3.1.2
23	C	Solve then interpret (curriculum exemplar)	B9.2.3.1.2
24	A	Solve then interpret (curriculum exemplar)	B9.2.3.1.2
25	D	Solve then interpret (curriculum exemplar)	B9.2.3.1.2
26	A	Variables on both sides (curriculum exemplar)	B9.2.3.1.2
27	B	Rational coefficients (curriculum exemplar)	B9.2.3.1.2
28	A	Describe a graph	B9.2.3.1.2
29	D	Translate 'at least'	B9.2.3.1.3
30	B	Translate 'at most'	B9.2.3.1.3
31	D	Money problem (at least)	B9.2.3.1.3
32	B	Translate a word statement	B9.2.3.1.3
33	C	Translate 'at least'	B9.2.3.1.3
34	A	Fare problem (at most)	B9.2.3.1.3
35	D	Marks problem (at least)	B9.2.3.1.3
36	B	Perimeter problem (at most)	B9.2.3.1.3

Q	Ans	Curriculum Area	Cognitive Level
37	C	Consecutive integers	B9.2.3.1.3
38	D	Identity inequality	B9.2.3.1.3
39	C	Combine two inequalities	B9.2.3.1.3
40	B	Integer solutions	B9.2.3.1.3

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.2.3.1.1: 16 (40%) | B9.2.3.1.2: 12 (30%) | B9.2.3.1.3: 12 (30%)

Ch11**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Angles at a point	B9.3.1.1
2	A	Angles on a straight line	B9.3.1.1
3	C	Vertically opposite angles	B9.3.1.1
4	B	Angles on a straight line	B9.3.1.1
5	A	Vertically opposite angles	B9.3.1.1
6	D	Angles at a point	B9.3.1.1
7	C	Corresponding angles	B9.3.1.1
8	D	Alternate angles	B9.3.1.1
9	B	Co-interior angles	B9.3.1.1
10	C	Corresponding angles	B9.3.1.1
11	A	Alternate / co-interior angles	B9.3.1.1
12	C	Complementary angles	B9.3.1.1
13	D	Name polygons	B9.3.1.1.1
14	A	Name polygons	B9.3.1.1.1
15	D	Angle sum of a triangle	B9.3.1.1.1
16	B	Polygon angle sum (curriculum exemplar)	B9.3.1.1.1
17	D	Polygon angle sum (curriculum exemplar)	B9.3.1.1.1
18	A	Polygon angle sum (curriculum exemplar)	B9.3.1.1.1
19	B	Polygon angle sum	B9.3.1.1.1
20	A	Find n from the angle sum	B9.3.1.1.1
21	D	Regular polygon interior angle	B9.3.1.1.1
22	A	Regular polygon interior angle	B9.3.1.1.1
23	B	Exterior angles	B9.3.1.1.1
24	D	Missing angle in a quadrilateral	B9.3.1.1.1
25	B	Exterior angle of a triangle	B9.3.1.1.1
26	C	Missing angles in a hexagon (curriculum ex...	B9.3.1.1.1
27	A	Congruent triangles	B9.3.1.1.2
28	C	Similar triangles	B9.3.1.1.2
29	B	Similar triangles ratio (curriculum exemplar)	B9.3.1.1.2
30	A	Similar triangles scale factor	B9.3.1.1.2
31	B	Congruence conditions	B9.3.1.1.2
32	D	Congruence conditions	B9.3.1.1.2
33	C	Perimeters of similar triangles	B9.3.1.1.2
34	D	Areas of similar triangles	B9.3.1.1.2
35	B	Clock angles	B9.3.1.1
36	C	Angles on a straight line	B9.3.1.1

Q	Ans	Curriculum Area	Cognitive Level
37	B	Angles at a point	B9.3.1.1
38	A	Isosceles triangle angles	B9.3.1.1.1
39	D	Exterior angles → number of sides	B9.3.1.1.1
40	C	Angle sum → polygon	B9.3.1.1.1

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.3.1.1: 15 (37%) | B9.3.1.1.1: 17 (42%) | B9.3.1.1.2: 8 (20%)

Ch12

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Incentre (angle-bisector concurrency)	B9.3.1.2.1
2	C	Incentre equidistant from sides	B9.3.1.2.1
3	A	Constructing the incentre	B9.3.1.2.1
4	C	Inscribed-circle radius	B9.3.1.2.1
5	B	Circumcentre (perpendicular-bisector concu...	B9.3.1.2.1
6	C	Circumcentre equidistant from vertices	B9.3.1.2.1
7	B	Circumscribed circle	B9.3.1.2.1
8	A	Locus: perpendicular bisector	B9.3.1.2.1
9	C	Locus: angle bisector	B9.3.1.2.1
10	D	Locating the incentre	B9.3.1.2.1
11	C	Locating the circumcentre	B9.3.1.2.1
12	D	Construction instruments	B9.3.1.2
13	B	Inscribed circle (Fig. 1)	B9.3.1.2.1
14	C	Circumscribed circle radii (Fig. 2)	B9.3.1.2.1
15	B	Circumcentre of a right triangle	B9.3.1.2.1
16	A	Circumradius from hypotenuse	B9.3.1.2.1
17	B	Circumcentre position by triangle type	B9.3.1.2.1
18	A	Equilateral triangle centres coincide	B9.3.1.2.1
19	B	Square: side to diagonal (curriculum exemp...	B9.3.1.2.2
20	D	Square from diagonal	B9.3.1.2.2
21	C	Square from diagonal: side length	B9.3.1.2.2
22	B	Data for a rectangle	B9.3.1.2.2
23	D	Rectangle: side and diagonal	B9.3.1.2.2
24	A	Rectangle diagonal (Pythagoras)	B9.3.1.2.2
25	D	Rhombus property	B9.3.1.2.2
26	A	Rhombus: diagonals to side	B9.3.1.2.2
27	B	Data for a parallelogram	B9.3.1.2.2
28	D	SSS determines a triangle uniquely	B9.3.1.2.2
29	B	Triangle inequality	B9.3.1.2.2
30	C	Triangle inequality: least side	B9.3.1.2.2
31	B	SAS construction	B9.3.1.2.2
32	A	ASA construction	B9.3.1.2.2
33	D	Regular hexagon construction	B9.3.1.2.2
34	A	Hexagon perimeter	B9.3.1.2.2
35	C	Compound plane shape (curriculum exemplar)	B9.3.1.2.2
36	A	Locus: circle	B9.3.1.2

Q	Ans	Curriculum Area	Cognitive Level
37	D	Locus: perpendicular bisector	B9.3.1.2
38	C	Bisecting sides for the circumcircle	B9.3.1.2
39	D	Perpendicular-bisector arcs (Fig. 4)	B9.3.1.2
40	A	Inscribed circle of a square	B9.3.1.2.2

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.3.1.2: 5 (12%) | B9.3.1.2.1: 17 (42%) | B9.3.1.2.2: 18 (45%)

Ch13

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Meaning of a prism	B9.3.2.1
2	B	Faces of a cuboid	B9.3.2.1.1
3	A	Vertices of a cuboid	B9.3.2.1.1
4	C	Edges of a cuboid (Fig. 1)	B9.3.2.1.1
5	D	Identifying a triangular prism (Fig. 1)	B9.3.2.1.1
6	C	Net of a cuboid	B9.3.2.1.1
7	D	Net of a triangular prism	B9.3.2.1.1
8	A	Identifying a solid from faces, edges, ver...	B9.3.2.1.1
9	C	Deriving the cuboid surface area	B9.3.2.1
10	B	Lateral surface of a prism	B9.3.2.1
11	C	Total edge length of a cuboid	B9.3.2.1
12	B	Lateral faces of an n-gonal prism	B9.3.2.1
13	A	Surface area of a cuboid	B9.3.2.1.2
14	D	Surface area of a cube	B9.3.2.1.2
15	C	Cube side from surface area	B9.3.2.1.2
16	D	Surface area of a cuboid	B9.3.2.1.2
17	A	Surface area from a net (Fig. 2)	B9.3.2.1.2
18	D	Open box without lid	B9.3.2.1.2
19	B	Missing height from surface area	B9.3.2.1.2
20	A	Surface area of a cuboid	B9.3.2.1.2
21	C	Effect of doubling the side	B9.3.2.1.2
22	B	Wrapping a gift box	B9.3.2.1.2
23	C	Painting walls	B9.3.2.1.2
24	B	Joined cubes	B9.3.2.1.2
25	C	Triangular-prism surface-area formula	B9.3.2.1.3
26	D	Right-triangle prism surface area	B9.3.2.1.3
27	B	Surface area from a net (Fig. 3)	B9.3.2.1.3
28	A	Isosceles-triangle prism surface area	B9.3.2.1.3
29	B	Right-triangle prism surface area	B9.3.2.1.3
30	A	Tent cloth (Fig. 4)	B9.3.2.1.3
31	D	Identifying a solid from its net	B9.3.2.1.1
32	B	Equilateral-triangle prism surface area	B9.3.2.1.3
33	C	Surface area from area and perimeter	B9.3.2.1.3
34	D	Triangular bases of the prism	B9.3.2.1.1
35	C	Faces of a triangular prism	B9.3.2.1.1
36	D	Units of surface area	B9.3.2.1

Q	Ans	Curriculum Area	Cognitive Level
37	A	Comparing surface areas	B9.3.2.1.2
38	B	Net of a pentagonal prism	B9.3.2.1
39	D	Cuboid with square base	B9.3.2.1.2
40	A	Cube surface area from edge total	B9.3.2.1.2

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.3.2.1: 7 (17%) | B9.3.2.1.1: 10 (25%) | B9.3.2.1.2: 15 (37%) | B9.3.2.1.3: 8 (20%)

Ch14

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Position vectors (Fig. 1)	B9.3.2.2
2	B	Position vectors (Fig. 1)	B9.3.2.2
3	D	Column vector notation	B9.3.2.2
4	B	Magnitude of a vector	B9.3.2.2
5	A	Vector between two points	B9.3.2.2
6	C	Negative of a vector	B9.3.2.2
7	A	Equal vectors	B9.3.2.2
8	D	Scalar multiplication	B9.3.2.2
9	C	Condition for parallel vectors	B9.3.2.2.1
10	B	Recognising parallel vectors	B9.3.2.2.1
11	C	Finding an unknown in parallel vectors	B9.3.2.2.1
12	B	Parallel vectors (curriculum exemplar)	B9.3.2.2.1
13	A	Condition for perpendicular vectors	B9.3.2.2.1
14	C	Perpendicular vectors (curriculum exemplar)	B9.3.2.2.1
15	A	Recognising perpendicular vectors	B9.3.2.2.1
16	D	Dot product formula	B9.3.2.2.1
17	C	Unknown in perpendicular vectors	B9.3.2.2.1
18	D	Parallel vectors with a scale factor	B9.3.2.2.1
19	B	Testing parallel vectors	B9.3.2.2.1
20	C	Testing parallel and perpendicular	B9.3.2.2.1
21	A	Parallel vectors	B9.3.2.2.1
22	C	Recognising perpendicular vectors	B9.3.2.2.1
23	A	Triangle law	B9.3.2.2.2
24	B	Triangle law on a grid (Fig. 2)	B9.3.2.2.2
25	A	Subtracting column vectors	B9.3.2.2.2
26	B	Vector between two points (curriculum exem...	B9.3.2.2.2
27	D	Magnitude of PQ	B9.3.2.2.2
28	A	Isosceles triangle from vectors	B9.3.2.2.2
29	B	Parallelogram law (Fig. 3)	B9.3.2.2.2
30	A	Adding column vectors	B9.3.2.2.2
31	D	Combined scalar multiplication and addition	B9.3.2.2.2
32	C	Combined operations on vectors	B9.3.2.2.2
33	A	Equal vectors in a parallelogram	B9.3.2.2.2
34	C	Parallelogram vertices (curriculum exemplar)	B9.3.2.2.2
35	D	Resultant of two journeys	B9.3.2.2.2
36	B	Magnitude of a resultant	B9.3.2.2.2

Q	Ans	Curriculum Area	Cognitive Level
37	D	Finding an endpoint from a vector	B9.3.2.2.2
38	C	Midpoint and vectors	B9.3.2.2.2
39	D	Zero vector	B9.3.2.2.2
40	B	Bearing direction as a vector	B9.3.2.2

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.3.2.2: 9 (22%) | B9.3.2.2.1: 14 (35%) | B9.3.2.2.2: 17 (42%)

Ch15

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Defining an enlargement	B9.3.3.1
2	C	Angles are invariant	B9.3.3.1
3	D	Lengths scale by $ k $	B9.3.3.1
4	A	Area scales by k^2	B9.3.3.1
5	D	Scale factor 1	B9.3.3.1
6	C	Scale factor -1	B9.3.3.1
7	D	Finding the centre	B9.3.3.1
8	B	Perimeter scales by $ k $	B9.3.3.1
9	C	Everyday enlargement	B9.3.3.1.1
10	B	Projector image	B9.3.3.1.1
11	A	Shadows as enlargement	B9.3.3.1.1
12	B	Non-examples of enlargement	B9.3.3.1.1
13	A	Photocopier percentages	B9.3.3.1.1
14	D	Zoom as enlargement	B9.3.3.1.1
15	A	Scale factor from lengths	B9.3.3.1.2
16	B	Fractional scale factor	B9.3.3.1.2
17	C	Negative scale factor	B9.3.3.1.2
18	B	Scale factor from a diagram (Fig. 1)	B9.3.3.1.2
19	D	Negative scale factor from a diagram (Fig. 2)	B9.3.3.1.2
20	B	Fractional scale factor from a diagram (Fi...	B9.3.3.1.2
21	D	Image of a point	B9.3.3.1.2
22	A	Image under negative k	B9.3.3.1.2
23	C	Enlargement about another centre	B9.3.3.1.2
24	D	Effect of a fractional k	B9.3.3.1.2
25	C	Effect of $k < -1$	B9.3.3.1.2
26	B	Coordinate rule	B9.3.3.1.2
27	D	Perimeter ratio	B9.3.3.1.2
28	A	Area under enlargement	B9.3.3.1.2
29	B	Scale factor from areas	B9.3.3.1.2
30	C	Centre of enlargement (Fig. 1)	B9.3.3.1.2
31	B	Congruent shapes	B9.3.3.1.3
32	C	Similar shapes	B9.3.3.1.3
33	A	Enlargement gives similar images	B9.3.3.1.3
34	C	When the image is congruent	B9.3.3.1.3
35	B	Area ratio of similar triangles	B9.3.3.1.3
36	A	Proportional sides	B9.3.3.1.3

Q	Ans	Curriculum Area	Cognitive Level
37	D	Always-similar pairs	B9.3.3.1.3
38	A	Not necessarily similar	B9.3.3.1.3
39	C	Area ratio from k	B9.3.3.1.3
40	D	Verifying similarity	B9.3.3.1.3

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.3.3.1: 8 (20%) | B9.3.3.1.1: 6 (15%) | B9.3.3.1.2: 16 (40%) | B9.3.3.1.3: 10 (25%)

Ch16

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Quantitative data	B9.4.1.1.1
2	A	Qualitative data	B9.4.1.1.1
3	C	Choosing a method: survey (curriculum exem...	B9.4.1.1.1
4	D	Choosing a method: mode of travel (curricu...	B9.4.1.1.1
5	A	Choosing a method: observation	B9.4.1.1.1
6	D	Choosing a method: experiment (curriculum ...	B9.4.1.1.1
7	A	Interview method	B9.4.1.1.1
8	B	Secondary data	B9.4.1.1.1
9	D	Tally marks	B9.4.1.1.2
10	A	Total frequency	B9.4.1.1.2
11	D	Mode of ungrouped data	B9.4.1.1.2
12	B	Mode of ungrouped data	B9.4.1.1.2
13	C	Range	B9.4.1.1.2
14	D	Median of an even number of values	B9.4.1.1.2
15	B	Reading a stem-and-leaf plot (Fig. 2)	B9.4.1.1.2
16	A	Counting leaves (Fig. 2)	B9.4.1.1.2
17	D	Bar chart features	B9.4.1.1.2
18	A	Line graphs	B9.4.1.1.2
19	B	Pie chart angles	B9.4.1.1.2
20	C	Pictographs	B9.4.1.1.2
21	B	Waffle diagrams	B9.4.1.1.2
22	D	Box-and-whisker plots	B9.4.1.1.2
23	B	Class width	B9.4.1.1.2
24	C	Modal class	B9.4.1.1.2
25	D	Frequency of a value	B9.4.1.1.2
26	C	Histograms for grouped data	B9.4.1.1.2
27	A	Why histogram bars touch	B9.4.1.1.2
28	C	Missing frequency	B9.4.1.1.2
29	D	Mode from a histogram	B9.4.1.1.3
30	A	Modal class from a histogram (Fig. 1)	B9.4.1.1.3
31	B	Total frequency from a histogram (Fig. 1)	B9.4.1.1.3
32	D	Reading a frequency (Fig. 1)	B9.4.1.1.3
33	B	Comparing frequencies (Fig. 1)	B9.4.1.1.3
34	C	Histogram vs bar chart	B9.4.1.1.3
35	A	Mode of grouped data	B9.4.1.1.3
36	C	Modal class (curriculum exemplar table)	B9.4.1.1.3

Q	Ans	Curriculum Area	Cognitive Level
37	B	Keeping the mode	B9.4.1.1.3
38	A	Median class from a histogram (Fig. 1)	B9.4.1.1.3
39	B	Identifying the modal class	B9.4.1.1.3
40	C	Missing bar frequency	B9.4.1.1.3

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.4.1.1.1: 8 (20%) | B9.4.1.1.2: 20 (50%) | B9.4.1.1.3: 12 (30%)

Ch17

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Leading questions (curriculum exemplar)	B9.4.1.2.1
2	B	Rewriting biased questions (curriculum exe...	B9.4.1.2.1
3	A	Biased samples (curriculum exemplar)	B9.4.1.2.1
4	C	Correcting a biased sample (curriculum exe...	B9.4.1.2.1
5	B	Sources of bias	B9.4.1.2.1
6	C	Cultural sensitivity	B9.4.1.2.1
7	A	Timing bias	B9.4.1.2.1
8	D	Privacy	B9.4.1.2.1
9	B	Mean of ungrouped data	B9.4.1.2.2
10	D	Mean of ungrouped data	B9.4.1.2.2
11	C	Median after sorting	B9.4.1.2.2
12	B	Mode	B9.4.1.2.2
13	D	Range	B9.4.1.2.2
14	A	Minimum and maximum (curriculum exemplar)	B9.4.1.2.2
15	C	Range of the data (curriculum exemplar)	B9.4.1.2.2
16	D	Mean from a frequency table	B9.4.1.2.2
17	B	Sum from the mean	B9.4.1.2.2
18	D	Missing value from the mean	B9.4.1.2.2
19	A	Median of an even number of values	B9.4.1.2.2
20	C	Mode from a chart (Fig. 1)	B9.4.1.2.2
21	A	Choosing a measure	B9.4.1.2.2
22	D	Properties of the mean	B9.4.1.2.2
23	A	Mean of a sequence	B9.4.1.2.2
24	B	Updating the mean	B9.4.1.2.2
25	C	Interpreting the mode (curriculum exemplar)	B9.4.1.2.2
26	A	Interpreting the median	B9.4.1.2.2
27	B	Total frequency (Fig. 1)	B9.4.1.2.2
28	A	Mean from a chart (Fig. 1)	B9.4.1.2.2
29	B	Effect of an extreme value on the mean (cu...	B9.4.1.2.3
30	D	Mean is sensitive	B9.4.1.2.3
31	C	Median is robust	B9.4.1.2.3
32	D	Median unchanged by an extreme	B9.4.1.2.3
33	C	Mode unchanged by a single extreme	B9.4.1.2.3
34	A	Comparing sensitivity	B9.4.1.2.3
35	C	New mean after adding a value	B9.4.1.2.3
36	D	Direction of the effect (curriculum exemplar)	B9.4.1.2.3

Q	Ans	Curriculum Area	Cognitive Level
37	B	Which measure survives	B9.4.1.2.3
38	A	Outliers	B9.4.1.2.3
39	D	Extreme salaries (curriculum-style interpr...	B9.4.1.2.3
40	B	Mean and median after an extreme	B9.4.1.2.3

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.4.1.2.1: 8 (20%) | B9.4.1.2.2: 20 (50%) | B9.4.1.2.3: 12 (30%)

Ch18

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Sample space of a coin	B9.4.2.1
2	C	Sample space of a die	B9.4.2.1
3	B	Two-coin sample space	B9.4.2.1
4	A	Probability scale	B9.4.2.1
5	B	Certain and impossible	B9.4.2.1
6	A	Complementary probability	B9.4.2.1
7	B	Probability formula	B9.4.2.1
8	C	Fraction-decimal-percentage	B9.4.2.1
9	A	Without replacement → dependent (curriculum...	B9.4.2.1.1
10	D	First-draw probability (curriculum exemplar)	B9.4.2.1.1
11	A	Second-draw probability after a red	B9.4.2.1.1
12	D	Dependence of the trials (curriculum exemp...	B9.4.2.1.1
13	B	Sample space of two dependent draws	B9.4.2.1.1
14	D	With replacement → independent	B9.4.2.1.1
15	B	Two blues without replacement	B9.4.2.1.1
16	C	Yellow then blue	B9.4.2.1.1
17	A	First ace from a deck (curriculum exemplar)	B9.4.2.1.1
18	D	Two aces without replacement (curriculum e...	B9.4.2.1.1
19	C	Ace then Jack (curriculum exemplar)	B9.4.2.1.1
20	D	Second ace probability	B9.4.2.1.1
21	C	Why the denominator changes	B9.4.2.1.1
22	B	Red then red	B9.4.2.1.1
23	C	Same-colour probability	B9.4.2.1.1
24	B	Multiplication rule for dependent events	B9.4.2.1.1
25	C	Fraction to percentage	B9.4.2.1.2
26	A	Decimal to fraction	B9.4.2.1.2
27	D	Fraction to decimal	B9.4.2.1.2
28	B	Percentage to fraction	B9.4.2.1.2
29	C	Probability as a ratio	B9.4.2.1.2
30	D	Tree-diagram path (Fig. 1)	B9.4.2.1.2
31	A	Branch probabilities sum to 1 (Fig. 1)	B9.4.2.1.2
32	D	Small probability as percentage	B9.4.2.1.2
33	B	Fraction to percentage	B9.4.2.1.2
34	A	Probability as a ratio	B9.4.2.1.2
35	D	Two greens without replacement	B9.4.2.1.2
36	C	At least one red via the complement	B9.4.2.1.2

Q	Ans	Curriculum Area	Cognitive Level
37	D	Fraction to percentage	B9.4.2.1.2
38	B	Fraction to decimal	B9.4.2.1.2
39	A	Small probability as percentage	B9.4.2.1.2
40	C	Decimal to ratio	B9.4.2.1.2

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.4.2.1: 8 (20%) | B9.4.2.1.1: 16 (40%) | B9.4.2.1.2: 16 (40%)

Ch1**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Decimal places & significant figures	Apply
2	B	Decimal places & significant figures	Apply
3	C	Decimal places & significant figures	Understand
4	A	Decimal places & significant figures	Apply
5	C	Decimal places & significant figures	Apply
6	B	Decimal places & significant figures	Apply
7	A	Decimal places & significant figures	Understand
8	D	Decimal places & significant figures	Apply
9	B	Decimal places & significant figures	Understand
10	D	Decimal places & significant figures	Apply
11	C	Decimal places & significant figures	Analyse
12	A	Decimal places & significant figures	Analyse
13	D	Decimal places & significant figures	Analyse
14	C	Decimal places & significant figures	Apply
15	A	Place value & standard form	Apply
16	B	Place value & standard form	Apply
17	A	Place value & standard form	Understand
18	C	Place value & standard form	Understand
19	D	Place value & standard form	Apply
20	B	Place value & standard form	Apply
21	C	Place value & standard form	Apply
22	A	Place value & standard form	Apply
23	B	Place value & standard form	Apply
24	D	Place value & standard form	Apply
25	C	Applications of rounding	Apply
26	A	Applications of rounding	Apply
27	B	Applications of rounding	Apply
28	D	Applications of rounding	Apply
29	C	Applications of rounding	Understand
30	A	Applications of rounding	Apply
31	B	Applications of rounding	Apply
32	D	Applications of rounding	Apply
33	C	Applications of rounding	Apply
34	B	Applications of rounding	Apply
35	A	Approximating square roots	Understand
36	D	Approximating square roots	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	A	Approximating square roots	Apply
38	C	Approximating square roots	Apply
39	D	Approximating square roots	Remember
40	B	Approximating square roots	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 29 (72%) | Remember: 1 (2%) | Understand: 7 (17%)

Ch2

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	The rational number system	Apply
2	D	The rational number system	Understand
3	A	The rational number system	Understand
4	B	The rational number system	Understand
5	D	The rational number system	Apply
6	B	The rational number system	Apply
7	A	The rational number system	Understand
8	C	The rational number system	Understand
9	D	The rational number system	Understand
10	B	The rational number system	Apply
11	C	The rational number system	Understand
12	A	The rational number system	Apply
13	C	Sets & two-set Venn diagrams	Apply
14	D	Sets & two-set Venn diagrams	Apply
15	A	Sets & two-set Venn diagrams	Apply
16	B	Sets & two-set Venn diagrams	Understand
17	C	Sets & two-set Venn diagrams	Understand
18	B	Sets & two-set Venn diagrams	Apply
19	D	Sets & two-set Venn diagrams	Understand
20	A	Sets & two-set Venn diagrams	Apply
21	C	Two-set real-life problems	Apply
22	B	Two-set real-life problems	Apply
23	D	Two-set real-life problems	Apply
24	A	Two-set real-life problems	Apply
25	C	Two-set real-life problems	Apply
26	B	Two-set real-life problems	Apply
27	D	Two-set real-life problems	Apply
28	A	Two-set real-life problems	Apply
29	D	Two-set real-life problems	Apply
30	B	Two-set real-life problems	Apply
31	C	Three-set problems	Apply
32	A	Three-set problems	Apply
33	B	Three-set problems	Apply
34	C	Three-set problems	Analyse
35	D	Three-set problems	Apply
36	A	Three-set problems	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Three-set problems	Apply
38	B	Three-set problems	Apply
39	A	Three-set problems	Understand
40	D	Three-set problems	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 1 (2%) | Apply: 27 (67%) | Understand: 12 (30%)

Ch3**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Powers of 10 & benchmark fractions	Apply
2	C	Powers of 10 & benchmark fractions	Apply
3	A	Powers of 10 & benchmark fractions	Apply
4	D	Powers of 10 & benchmark fractions	Apply
5	B	Powers of 10 & benchmark fractions	Understand
6	A	Powers of 10 & benchmark fractions	Apply
7	D	Powers of 10 & benchmark fractions	Apply
8	C	Powers of 10 & benchmark fractions	Apply
9	D	Powers of 10 & benchmark fractions	Apply
10	A	Powers of 10 & benchmark fractions	Understand
11	C	Commutative property	Understand
12	B	Commutative property	Apply
13	D	Commutative property	Understand
14	C	Commutative property	Understand
15	B	Commutative property	Understand
16	A	Commutative property	Understand
17	B	Associative property	Understand
18	A	Associative property	Apply
19	C	Associative property	Understand
20	D	Associative property	Apply
21	A	Associative property	Understand
22	C	Associative property	Apply
23	D	Distributive property	Apply
24	B	Distributive property	Apply
25	D	Distributive property	Apply
26	C	Distributive property	Apply
27	B	Distributive property	Apply
28	A	Distributive property	Apply
29	B	Distributive property	Understand
30	A	Distributive property	Apply
31	D	Mental mathematics applications	Apply
32	C	Mental mathematics applications	Apply
33	B	Mental mathematics applications	Apply
34	C	Mental mathematics applications	Apply
35	A	Mental mathematics applications	Apply
36	D	Mental mathematics applications	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	A	Mental mathematics applications	Apply
38	D	Mental mathematics applications	Apply
39	C	Mental mathematics applications	Apply
40	B	Mental mathematics applications	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Apply: 29 (72%) | Understand: 11 (27%)

Ch4

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Adding & subtracting decimals	Apply
2	A	Adding & subtracting decimals	Apply
3	C	Adding & subtracting decimals	Understand
4	B	Adding & subtracting decimals	Apply
5	D	Adding & subtracting decimals	Apply
6	C	Adding & subtracting decimals	Apply
7	B	Adding & subtracting decimals	Apply
8	A	Adding & subtracting decimals	Understand
9	C	Adding & subtracting decimals	Apply
10	B	Adding & subtracting decimals	Understand
11	D	Multiplying & dividing decimals	Apply
12	A	Multiplying & dividing decimals	Apply
13	B	Multiplying & dividing decimals	Apply
14	D	Multiplying & dividing decimals	Apply
15	C	Multiplying & dividing decimals	Apply
16	A	Multiplying & dividing decimals	Apply
17	B	Multiplying & dividing decimals	Apply
18	C	Multiplying & dividing decimals	Apply
19	D	Multiplying & dividing decimals	Apply
20	A	Multiplying & dividing decimals	Apply
21	D	Rounding answers	Apply
22	C	Rounding answers	Apply
23	A	Rounding answers	Apply
24	B	Rounding answers	Apply
25	C	Rounding answers	Apply
26	B	Rounding answers	Apply
27	D	Rounding answers	Apply
28	A	Rounding answers	Apply
29	C	Word problems	Apply
30	B	Word problems	Analyse
31	D	Word problems	Apply
32	A	Word problems	Analyse
33	B	Word problems	Understand
34	C	Word problems	Apply
35	A	Word problems	Apply
36	D	Word problems	Analyse

Q	Ans	Curriculum Area	Cognitive Level
37	A	Word problems	Apply
38	C	Word problems	Apply
39	D	Word problems	Apply
40	B	Word problems	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 33 (82%) | Understand: 4 (10%)

Ch5

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Identify a surd	B9.1.2.4.1
2	D	Recognise a perfect square	B9.1.2.4.1
3	C	Identify a compound surd (curriculum exemp...	B9.1.2.4.1
4	B	Identify like surds	B9.1.2.4.1
5	A	Concept of surds	B9.1.2.4.1
6	B	Recognise a perfect square	B9.1.2.4.1
7	A	Simplify a surd	B9.1.2.4.3
8	D	Simplify a surd	B9.1.2.4.3
9	B	Simplify a surd	B9.1.2.4.3
10	D	Simplify a surd (curriculum exemplar)	B9.1.2.4.3
11	A	Simplify a surd	B9.1.2.4.3
12	D	Simplify a surd	B9.1.2.4.3
13	C	Simplify a surd	B9.1.2.4.3
14	D	Simplify a surd (curriculum exemplar)	B9.1.2.4.3
15	C	Simplify a surd	B9.1.2.4.3
16	A	Simplify a surd	B9.1.2.4.3
17	C	Add like surds	B9.1.2.4.2
18	A	Subtract like surds	B9.1.2.4.2
19	B	Add like surds	B9.1.2.4.2
20	C	Add like surds	B9.1.2.4.2
21	B	Add like surds	B9.1.2.4.2
22	C	Combine like surds	B9.1.2.4.2
23	B	Subtract like surds	B9.1.2.4.2
24	D	Recognise equivalent surd forms	B9.1.2.4.1
25	A	Simplify, then add like surds	B9.1.2.4.3
26	C	Add like surds	B9.1.2.4.2
27	B	Add like surds	B9.1.2.4.2
28	D	Add like surds	B9.1.2.4.2
29	C	Simplify, then add like surds	B9.1.2.4.3
30	D	Unlike surds cannot be combined	B9.1.2.4.2
31	A	Multiply surds	B9.1.2.4.2
32	D	Multiply surds	B9.1.2.4.2
33	A	Multiply surds	B9.1.2.4.2
34	B	Multiply surds	B9.1.2.4.2
35	D	Multiply surds	B9.1.2.4.2
36	B	Multiply surds	B9.1.2.4.2

Q	Ans	Curriculum Area	Cognitive Level
37	C	Divide surds	B9.1.2.4.2
38	A	Divide surds	B9.1.2.4.2
39	B	Divide surds	B9.1.2.4.2
40	A	Divide surds	B9.1.2.4.2

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.1.2.4.1: 7 (17%) | B9.1.2.4.2: 21 (52%) | B9.1.2.4.3: 12 (30%)

Ch6

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Equivalent fractions	B9.1.3.1.1
2	A	Simplest form (curriculum exemplar)	B9.1.3.1.1
3	D	Improper fraction to mixed number (curricu...	B9.1.3.1.1
4	A	Mixed number to improper fraction (curricu...	B9.1.3.1.1
5	B	Compare fractions	B9.1.3.1.1
6	D	Order fractions	B9.1.3.1.1
7	C	Lowest terms	B9.1.3.1.1
8	B	Decimal to fraction	B9.1.3.1.1
9	A	Add fractions (curriculum exemplar)	B9.1.3.1.1
10	D	Add fractions	B9.1.3.1.1
11	C	Subtract fractions	B9.1.3.1.1
12	B	Mixed operations on fractions (curriculum ...	B9.1.3.1.1
13	C	Subtract mixed numbers	B9.1.3.1.1
14	A	Add fractions	B9.1.3.1.1
15	C	Subtract fractions	B9.1.3.1.1
16	B	Complements of fractions	B9.1.3.1.1
17	C	Multiply fractions (curriculum exemplar)	B9.1.3.1.1
18	D	Divide fractions	B9.1.3.1.1
19	B	Fraction of a quantity	B9.1.3.1.1
20	A	Multiply fractions	B9.1.3.1.1
21	C	Divide by a unit fraction	B9.1.3.1.1
22	B	Square of a fraction	B9.1.3.1.1
23	D	Divide a fraction by a whole number	B9.1.3.1.1
24	A	Divide fractions, then add (curriculum exe...	B9.1.3.1.1
25	B	BODMAS with fractions (curriculum exemplar)	B9.1.3.1.2
26	C	BODMAS with fractions (curriculum exemplar)	B9.1.3.1.2
27	D	BODMAS with fractions (curriculum exemplar)	B9.1.3.1.2
28	C	Order of operations	B9.1.3.1.2
29	A	Order of operations with brackets	B9.1.3.1.2
30	B	Brackets first	B9.1.3.1.2
31	D	Order of operations	B9.1.3.1.2
32	C	Successive fractions of a quantity	B9.1.3.1.2
33	A	Percentages as fractions (curriculum exemp...	B9.1.3.1.3
34	B	Percentage reduction (curriculum exemplar)	B9.1.3.1.3
35	C	Checking claims with fractions (curriculum...	B9.1.3.1.3
36	A	Comparing fractions in context (curriculum...	B9.1.3.1.3

Q	Ans	Curriculum Area	Cognitive Level
37	D	Fractions of quantities in context	B9.1.3.1.3
38	A	Division of fractions in context	B9.1.3.1.3
39	D	Fractions of quantities in context	B9.1.3.1.3
40	B	Rounding results to decimal places	B9.1.3.1.3

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.1.3.1.1: 24 (60%) | B9.1.3.1.2: 8 (20%) | B9.1.3.1.3: 8 (20%)

Ch7

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Simplify a ratio	B9.1.4.1.2
2	A	Share in a given ratio	B9.1.4.1.2
3	B	Find quantities from a ratio	B9.1.4.1.2
4	D	Speed as a rate	B9.1.4.1.4
5	B	Cost through unit rate	B9.1.4.1.4
6	A	Proportional equation $t = pn$	B9.1.4.1.1
7	C	Recognise direct proportion	B9.1.4.1.4
8	B	Read a proportional graph (curriculum exem...	B9.1.4.1.4
9	D	Use a proportional graph (curriculum exemp...	B9.1.4.1.4
10	A	Unit rate as slope (curriculum exemplar)	B9.1.4.1.4
11	D	Simple interest formula	B9.1.4.1.2
12	B	Simple interest (curriculum exemplar)	B9.1.4.1.2
13	A	Amount after interest (curriculum exemplar)	B9.1.4.1.2
14	D	Amount after extended period (curriculum e...	B9.1.4.1.2
15	C	Amount after interest	B9.1.4.1.2
16	B	Find the principal	B9.1.4.1.2
17	D	VAT calculation	B9.1.4.1.2
18	B	VAT-inclusive price (curriculum exemplar)	B9.1.4.1.2
19	D	VAT paid (curriculum exemplar)	B9.1.4.1.2
20	A	Discount (curriculum exemplar)	B9.1.4.1.2
21	C	Discount amount (curriculum exemplar)	B9.1.4.1.2
22	D	Commission (curriculum exemplar)	B9.1.4.1.2
23	B	Depreciation, first year (curriculum exemp...	B9.1.4.1.2
24	C	Depreciation, second year (curriculum exem...	B9.1.4.1.2
25	A	NHIL exclusive price (curriculum exemplar)	B9.1.4.1.2
26	C	NHIL charged (curriculum exemplar)	B9.1.4.1.2
27	D	Insurance premium and value (curriculum ex...	B9.1.4.1.2
28	C	SSNIT worker rate, Act 766 (curriculum exe...	B9.1.4.1.3
29	D	SSNIT employer rate, Act 766 (curriculum e...	B9.1.4.1.3
30	A	SSNIT total rate, Act 766 (curriculum exem...	B9.1.4.1.3
31	C	Worker contribution, Act 766 (curriculum e...	B9.1.4.1.3
32	B	Employer contribution, Act 766 (curriculum...	B9.1.4.1.3
33	C	Total SSNIT, Act 766 (curriculum exemplar)	B9.1.4.1.3
34	B	SSNIT worker rate, PNDCL 247 (curriculum e...	B9.1.4.1.3
35	C	Employer contribution, PNDCL 247 (curricul...	B9.1.4.1.3
36	A	Basic salary from contribution (curriculum...	B9.1.4.1.3

Q	Ans	Curriculum Area	Cognitive Level
37	D	Ratio from an equation	B9.1.4.1.2
38	A	Map scale as a rate	B9.1.4.1.4
39	C	Inverse proportion	B9.1.4.1.4
40	A	Constant of proportionality	B9.1.4.1.1

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.1.4.1.1: 2 (5%) | B9.1.4.1.2: 21 (52%) | B9.1.4.1.3: 9 (22%) | B9.1.4.1.4: 8 (20%)

Ch8

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Table of values	B9.2.1.1.1
2	A	Table of values (curriculum exemplar)	B9.2.1.1.1
3	C	Table of values (curriculum exemplar)	B9.2.1.1.1
4	A	Table of values	B9.2.1.1.1
5	B	Ordered pairs on a line	B9.2.1.1.1
6	A	Points on a line	B9.2.1.1.1
7	C	Complete a table	B9.2.1.1.1
8	B	Complete a table	B9.2.1.1.1
9	C	Gradient	B9.2.1.1.2
10	D	y-intercept	B9.2.1.1.2
11	C	Ordered pair (curriculum exemplar)	B9.2.1.1.2
12	B	Ordered pair (curriculum exemplar)	B9.2.1.1.2
13	C	Intersection of two lines (curriculum exem...	B9.2.1.1.2
14	B	x-intercept	B9.2.1.1.2
15	A	y-intercept	B9.2.1.1.2
16	C	Gradient	B9.2.1.1.2
17	B	Simultaneous equations (curriculum exemplar)	B9.2.1.1.3
18	D	Simultaneous equations (curriculum exemplar)	B9.2.1.1.3
19	B	Simultaneous equations	B9.2.1.1.3
20	C	Simultaneous equations	B9.2.1.1.3
21	A	Parallel lines: no solution	B9.2.1.1.3
22	D	Coincident lines	B9.2.1.1.3
23	B	Read the intersection (Fig. 1)	B9.2.1.1.3
24	A	Read a value from the graph	B9.2.1.1.2
25	B	Elimination method	B9.2.1.1.3
26	D	Elimination method	B9.2.1.1.3
27	B	Elimination method	B9.2.1.1.3
28	D	Word problem	B9.2.1.1.3
29	A	Word problem (cost)	B9.2.1.1.3
30	D	Word problem (ages)	B9.2.1.1.3
31	C	Word problem (livestock)	B9.2.1.1.3
32	D	Substitution method	B9.2.1.1.3
33	A	Table of values	B9.2.1.1.1
34	D	Table of values	B9.2.1.1.1
35	A	Relation from a table	B9.2.1.1.1
36	C	Table from a relation	B9.2.1.1.1

Q	Ans	Curriculum Area	Cognitive Level
37	A	Shape of a linear graph	B9.2.1.1.1
38	B	x-intercept	B9.2.1.1.2
39	D	Solution from intersection	B9.2.1.1.3
40	C	Graphical method	B9.2.1.1.3

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.2.1.1.1: 13 (32%) | B9.2.1.1.2: 10 (25%) | B9.2.1.1.3: 17 (42%)

Ch9

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Change of subject (simple)	B9.2.2.1.1
2	C	Change of subject (simple)	B9.2.2.1.1
3	A	Change of subject (circumference)	B9.2.2.1.1
4	D	Change of subject (triangle area)	B9.2.2.1.1
5	A	Change of subject (motion)	B9.2.2.1.1
6	B	Change of subject (rectangle)	B9.2.2.1.1
7	C	Change of subject (two steps)	B9.2.2.1.1
8	B	Change of subject (square root)	B9.2.2.1.1
9	A	Change of subject (perimeter)	B9.2.2.1.1
10	C	Change of subject (squares)	B9.2.2.1.1
11	D	Use a changed subject (rectangle, curriculum...)	B9.2.2.1.1
12	B	Use a changed subject (circle, curriculum ...)	B9.2.2.1.1
13	C	Use a changed subject (triangle, curriculum...)	B9.2.2.1.1
14	B	Use a changed subject (cylinder, curriculum...)	B9.2.2.1.1
15	D	Substitute into an expression	B9.2.2.1.2
16	A	Substitute two values	B9.2.2.1.2
17	D	Substitute with a negative value (curriculum...)	B9.2.2.1.2
18	A	Substitute a negative value	B9.2.2.1.2
19	B	Substitute into brackets	B9.2.2.1.2
20	D	Substitute into a formula	B9.2.2.1.2
21	A	Substitute into a volume formula (curriculum...)	B9.2.2.1.2
22	D	Substitute into the SI formula	B9.2.2.1.2
23	B	Substitute into a motion formula	B9.2.2.1.2
24	D	Substitute into squares	B9.2.2.1.2
25	C	Substitute into a fraction	B9.2.2.1.2
26	A	Substitute under a root	B9.2.2.1.2
27	D	Common factor (curriculum exemplar)	B9.2.2.1.3
28	C	Common factor (curriculum exemplar)	B9.2.2.1.3
29	B	Negative common factor (curriculum exemplar)	B9.2.2.1.3
30	C	Common factor (curriculum exemplar)	B9.2.2.1.3
31	A	Numerical common factor	B9.2.2.1.3
32	C	Variable common factor	B9.2.2.1.3
33	B	Common factor with powers	B9.2.2.1.3
34	D	Difference of two squares	B9.2.2.1.3
35	A	Factorisation by grouping	B9.2.2.1.3
36	D	Quadratic trinomial	B9.2.2.1.3

Q	Ans	Curriculum Area	Cognitive Level
37	A	Numerical common factor	B9.2.2.1.3
38	C	Difference of two squares	B9.2.2.1.3
39	B	Common factor of terms	B9.2.2.1.3
40	C	Common factor with two variables	B9.2.2.1.3

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

B9.2.2.1.1: 14 (35%) | B9.2.2.1.2: 12 (30%) | B9.2.2.1.3: 14 (35%)

Mathematics Formula Sheet — Basic 9

Number and Financial Mathematics

Simple Interest	$I = PRT / 100$
Amount (Simple Interest)	$A = P + I = P(1 + RT/100)$
Compound Interest	$A = P(1 + R/100)^n$
Depreciation	$V = P(1 - R/100)^n$
Profit	Profit = Selling Price - Cost Price
Loss	Loss = Cost Price - Selling Price
Profit %	Profit% = (Profit/Cost Price) $\times$ 100
Loss %	Loss% = (Loss/Cost Price) $\times$ 100
Discount	Discount = Marked Price - Selling Price
Discount %	Discount% = (Discount/Marked Price) $\times$ 100
Commission	Commission = Rate $\times$ Sales Amount
Tax	Tax = Rate $\times$ Taxable Income
Exchange Rate	Amount in new currency = Amount $\times$ Exchange Rate

Algebra

Quadratic Equation	$ax^2 + bx + c = 0$
Quadratic Formula	$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$
Factorising Quadratics	$ax^2 + bx + c = a(x-p)(x-q)$
Completing the Square	$ax^2 + bx + c = a(x + b/2a)^2 + (c - b^2/4a)$
Discriminant	$\Delta = b^2 - 4ac$
Nature of Roots	$\Delta > 0$: two real roots; $\Delta = 0$: equal roots; $\Delta < 0$: no real roots
Sum of Roots	$\alpha + \beta = -b/a$
Product of Roots	$\alpha \times \beta = c/a$
Laws of Indices	$a^m \times a^n = a^{m+n}$; $a^m/a^n = a^{m-n}$; $(a^m)^n = a^{mn}$
Negative Index	$a^{-n} = 1/a^n$
Fractional Index	$a^{(m/n)} = \sqrt[n]{a^m}$
Zero Index	$a^0 = 1$
Variation (Direct)	$y \propto x$ means $y = kx$
Variation (Inverse)	$y \propto 1/x$ means $y = k/x$
Joint Variation	$y \propto xz$ means $y = kxz$

Geometry and Trigonometry

Sine Rule	$a/\sin A = b/\sin B = c/\sin C$
Cosine Rule	$a^2 = b^2 + c^2 - 2bc \cdot \cos A$
Area of Triangle (Trig)	$\text{Area} = \frac{1}{2}ab \cdot \sin C$
Arc Length	$s = r\theta$ (θ in radians)
Area of Sector	$A = \frac{1}{2}r^2\theta$ (θ in radians)
Radians to Degrees	$\theta^\circ = \theta(\text{rad}) \times 180/\pi$
Degrees to Radians	$\theta(\text{rad}) = \theta^\circ \times \pi/180$
Bearing	Angle measured clockwise from north (3 figures)
Angle of Elevation	Angle measured upward from horizontal
Angle of Depression	Angle measured downward from horizontal

Coordinate Geometry

Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$
Gradient	$m = (y_2 - y_1)/(x_2 - x_1)$
Equation of Line	$y - y_1 = m(x - x_1)$
Parallel Lines	Same gradient: $m_1 = m_2$
Perpendicular Lines	Product of gradients = -1: $m_1 \times m_2 = -1$

Statistics and Probability

Mean (Ungrouped)	$\bar{x} = \Sigma x / n$
Mean (Grouped)	$\bar{x} = \Sigma fx / \Sigma f$
Variance	$\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f$
Standard Deviation	$\sigma = \sqrt{\text{Variance}}$
Cumulative Frequency	Running total of frequencies
Interquartile Range	$\text{IQR} = Q_3 - Q_1$
Probability	$P(A) = n(A)/n(S)$
Mutually Exclusive	$P(A \text{ or } B) = P(A) + P(B)$
Independent Events	$P(A \text{ and } B) = P(A) \times P(B)$
Tree Diagram	Multiply along branches, add between branches

Sets

Union	$A \cup B = \text{elements in } A \text{ or } B \text{ or both}$
Intersection	$A \cap B = \text{elements in both } A \text{ and } B$
Complement	$A' = \text{elements not in } A$
$n(A \cup B)$	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Subset	$A \subset B$ means all elements of A are in B

Curriculum Reference

This book aligns with the following curriculum standards:

- Ghana Education Service (GES)
Standards-Based Curriculum for Junior High Schools (2019)
- National Council for Curriculum and Assessment (NaCCA)
Curriculum Framework for Basic Education
- West African Examinations Council (WAEC)
BECE Syllabus for Integrated Science / Mathematics

The content has been reviewed to ensure alignment with:

- Strand and sub-strand organization
- Learning indicators and objectives
- Cognitive levels (Knowledge, Understanding, Application, Analysis)
- Core competencies (Critical Thinking, Problem Solving, Communication)

Teachers and students should use this book in conjunction with the official curriculum documents available from the Ghana Education Service.

For more information:

Ghana Education Service: www.ges.gov.gh

NaCCA: www.nacca.gov.gh

WAEC: www.waec.org

About Beacon Educational Consult

Beacon Educational Consult is a Ghanaian educational publishing company dedicated to producing high-quality learning materials for Junior High School students across Ghana.

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